SND Mobile Design Workshop

Jeremy Gilbert & **Dave Stanton**

Northwestern University,
Evanston, IL 2011

About Us

Our Plan

	Today
9 AM	Registration
9:30 AM	Introductions
10 AM	UI Gestures & IA
Noon	Lunch
1 PM	The Range of Apps
2 PM	Mobile Frameworks

	Saturday
9 AM	Planning Your App
10 AM	Start
Noon	Lunch
1 PM	FP Work Time
4 PM	Wrapping Up

About You

- Your Name
- Your Smartphone (if you have one)
- Familiarity with HTML/CSS/Javascript:
    a. I read about HTML5 for fun
    b. I built my last site in tables
    c. What's this HTML thing?
- Where are you working/What do you do?

Mobile Design Bootcamp Goals

What do you want to learn?

What are we planning to cover

Smartphone Market

US Mobile Market



82%

of Americans use
a mobile phone

Pew Internet



17%

of Americans use
smartphones

NielsenWire: Global Mobile

Changing Habits

Newspapers

During Breakfast

Radio

Commuting

Television

After Dinner

Changing Habits

Multiple Sources & Applications



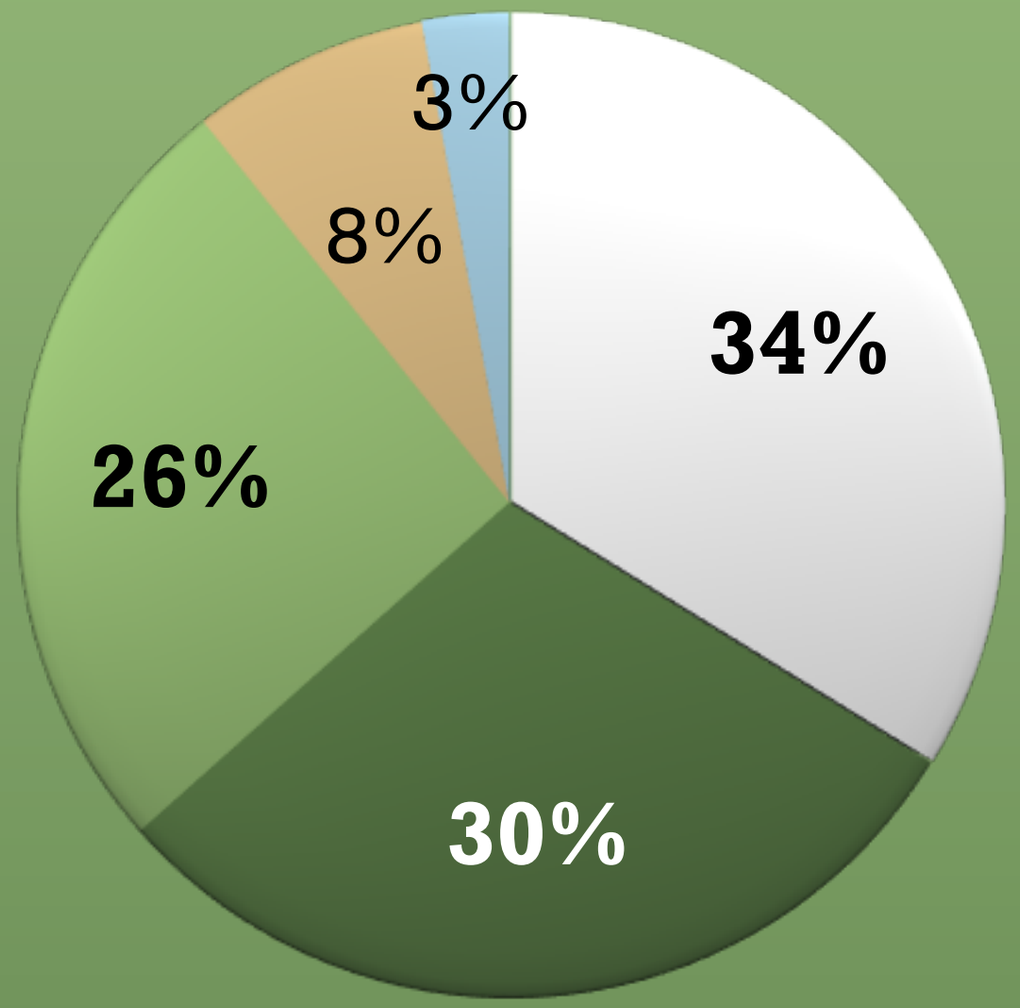
During Breakfast

Commuting

After Dinner

US Smartphones

- Google
- RIM
- Apple
- Microsoft
- HP



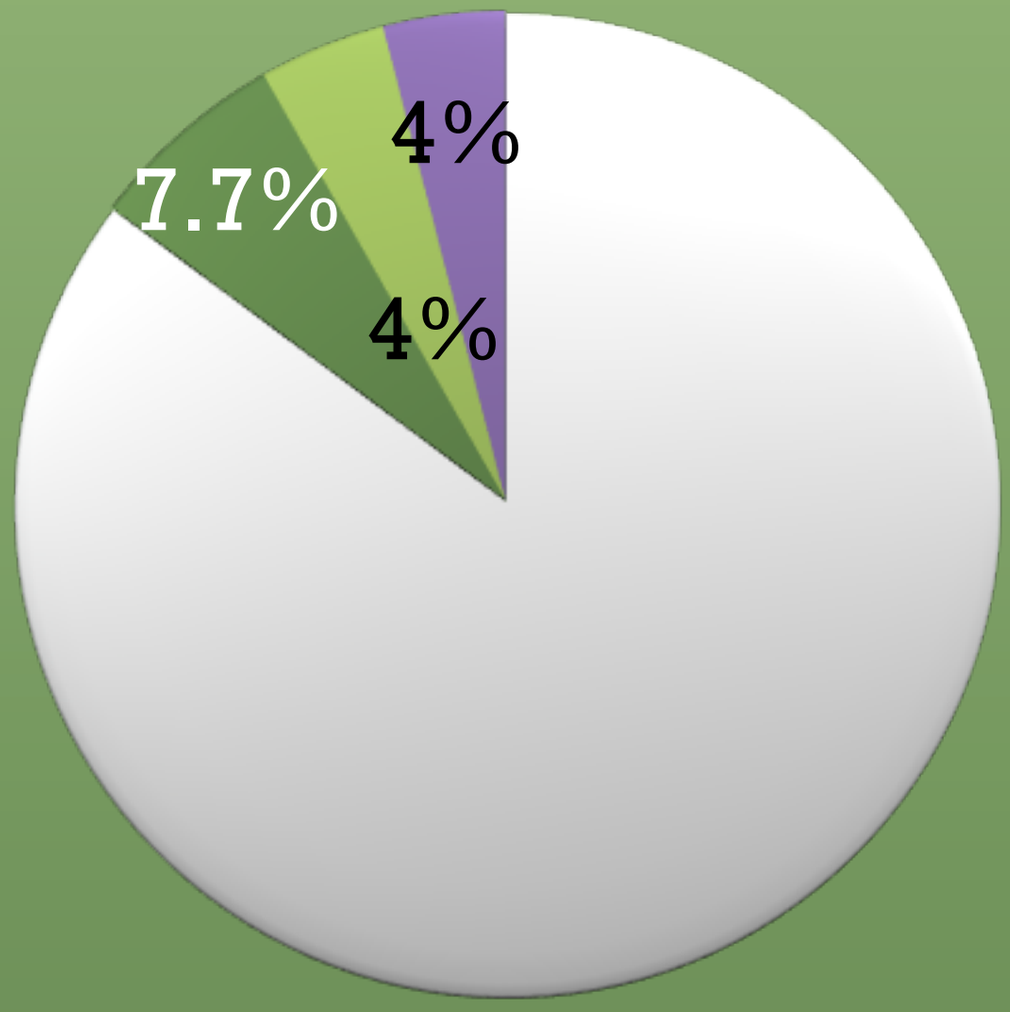
Smartphones Market Share By Operating System

Wired Gadget Lab | February 2011

Marketshare

82.7%

of smartphone apps sold are by Apple



- Apple
- RIM
- Nokia
- Google

IHS iSuppli | February 2011

Apple Has Sold More Than

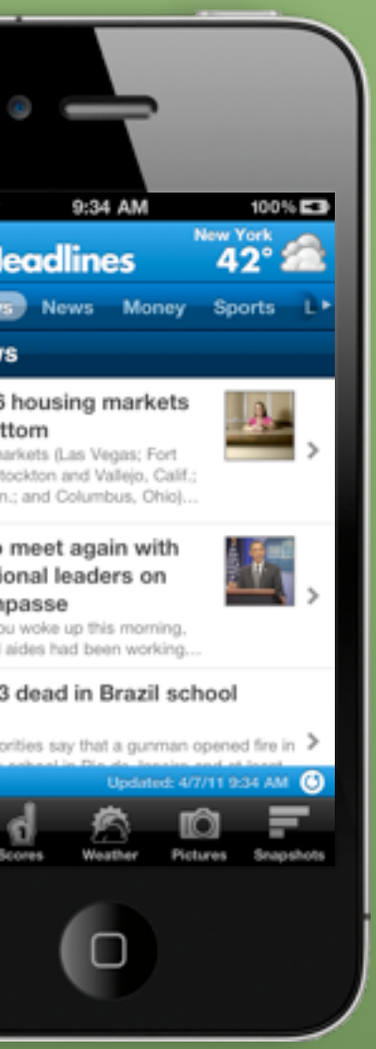
10 Billion

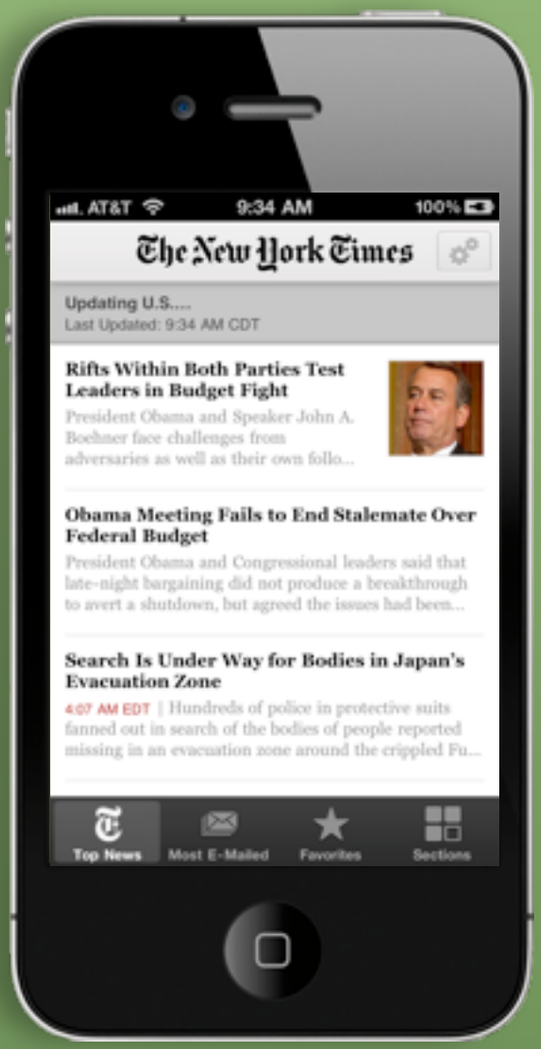
Mobile Applications

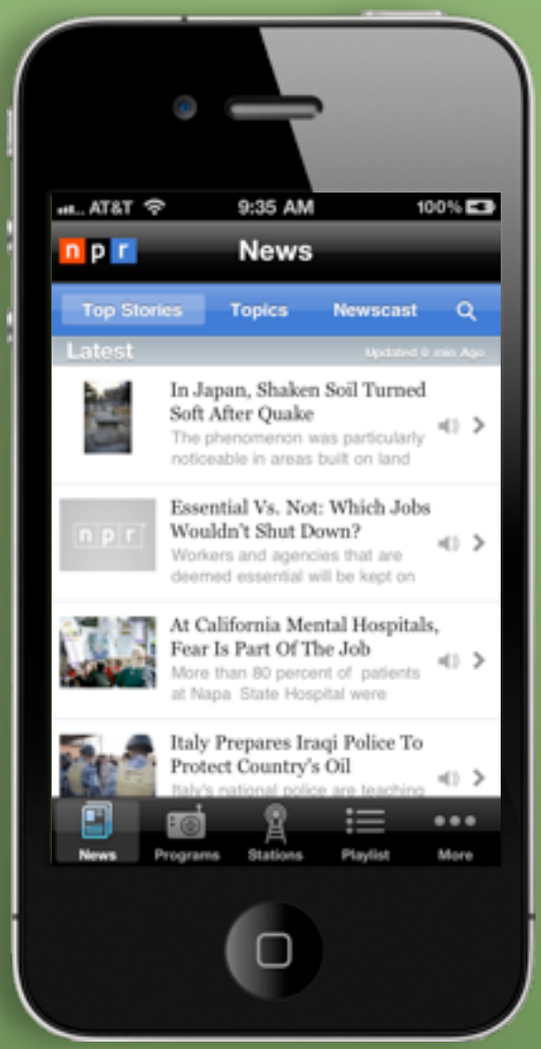
Apple | January 2011

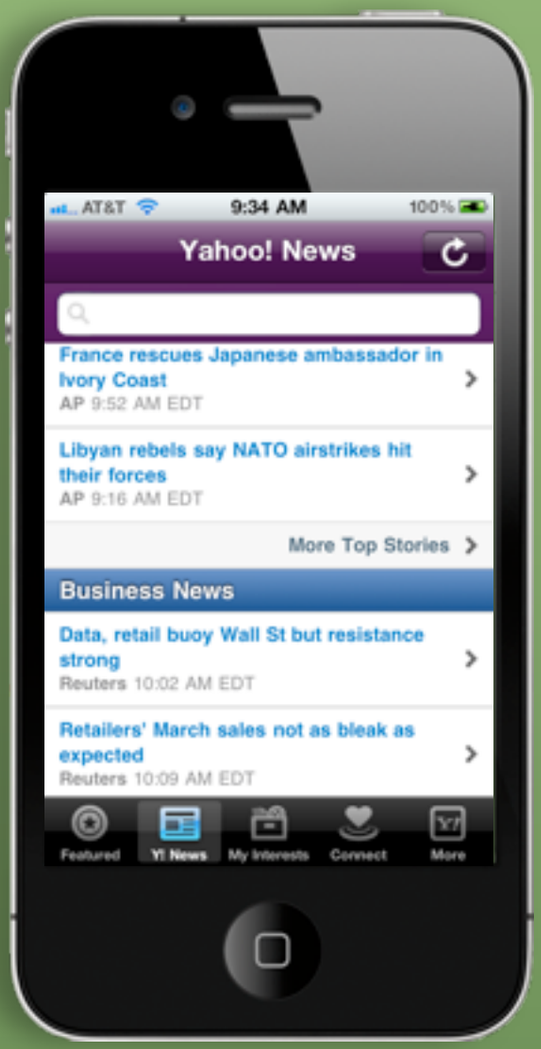
Ten Traits of Great Mobile Apps

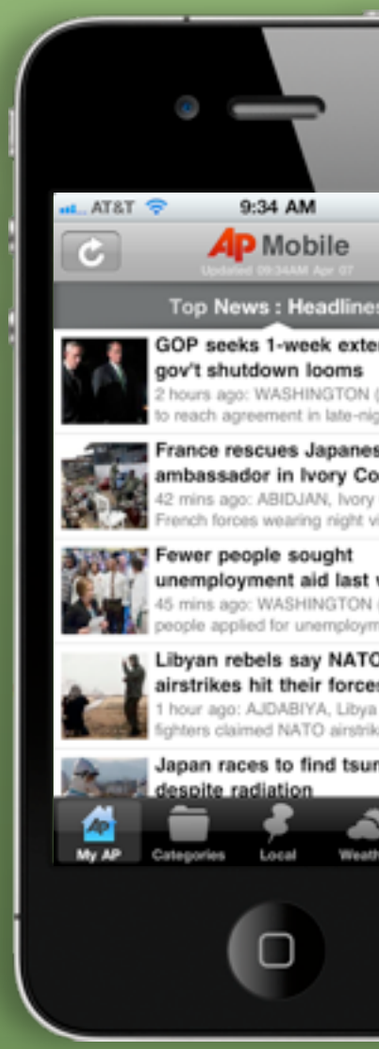
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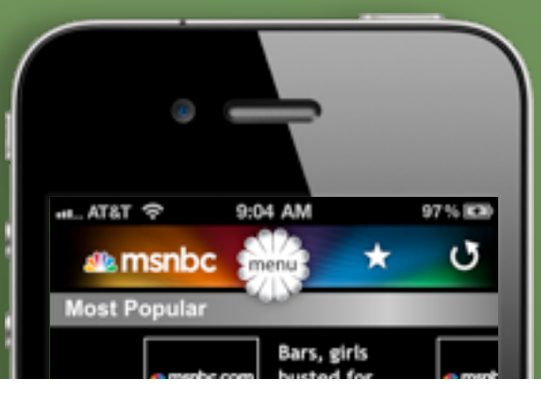

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unemployment aid last
Libyan rebels say NATO
airstrikes hit their force
Japan races to find tsun
despite radiation
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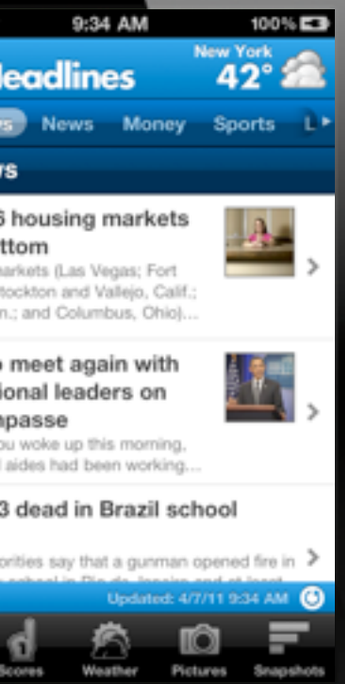

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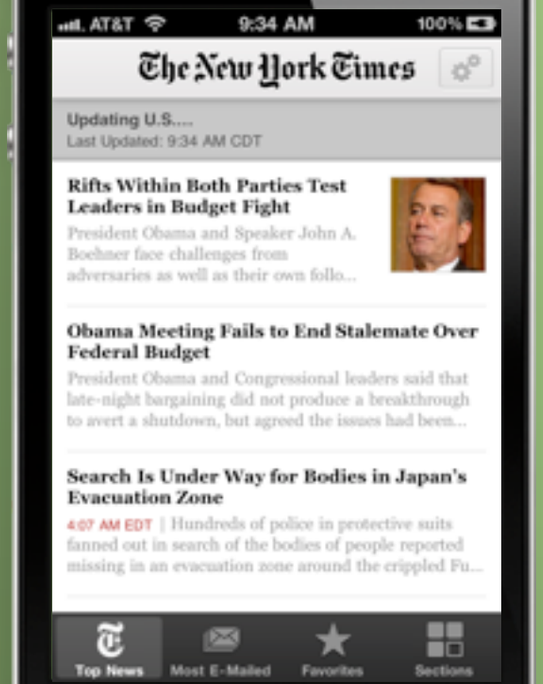

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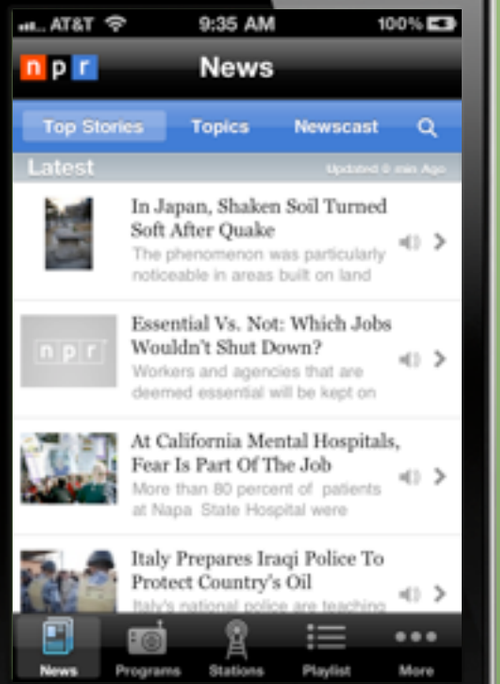

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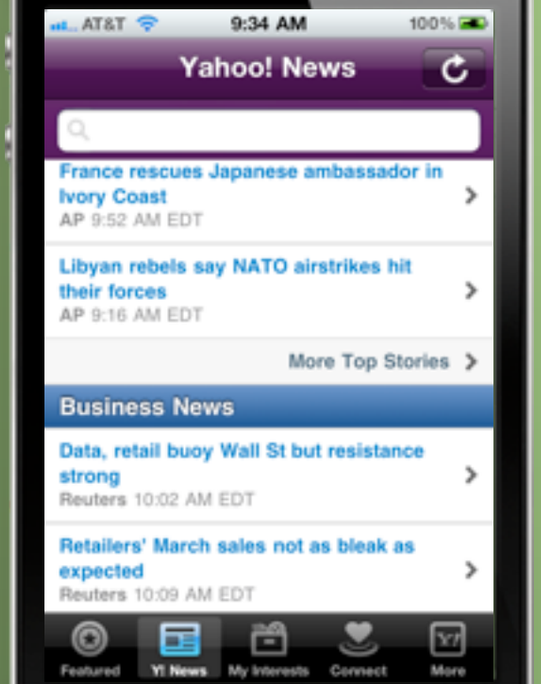

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Most Popular
Bars, girls

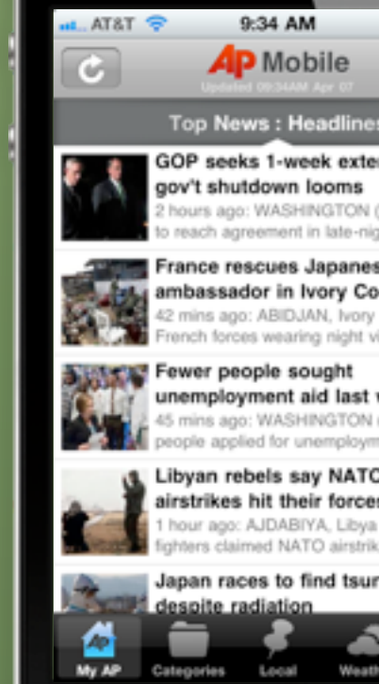

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The Washington Post
TOP STORIES
Obama says shutdown would

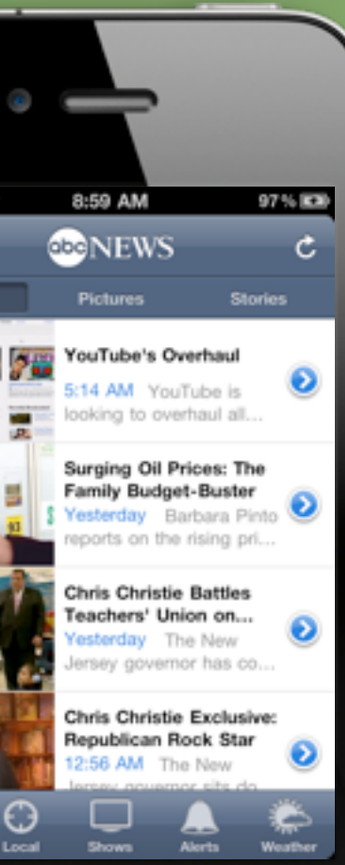

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Rifts Within Both Parties Test Leaders in Budget Fight
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Obama Meeting Fails to End Stalemate Over Federal Budget
President Obama and Congressional leaders said that late-night bargaining did not produce a breakthrough to avert a shutdown, but agreed the issues had been...
Search Is Under Way for Bodies in Japan's Evacuation Zone
4:07 AM EDT | Hundreds of police in protective suits fanned out in search of the bodies of people reported missing in an evacuation zone around the crippled Fu...
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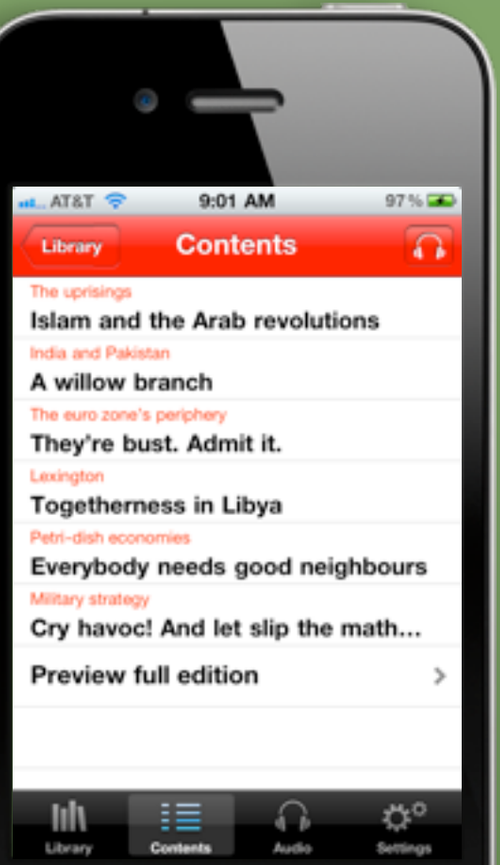

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In Japan, Shaken Soil Turned Soft After Quake
The phenomenon was particularly noticeable in areas built on land
Essential Vs. Not: Which Jobs Wouldn't Shut Down?
Workers and agencies that are deemed essential will be kept on
At California Mental Hospitals, Fear Is Part Of The Job
More than 80 percent of patients at Napa State Hospital were
Italy Prepares Iraqi Police To Protect Country's Oil
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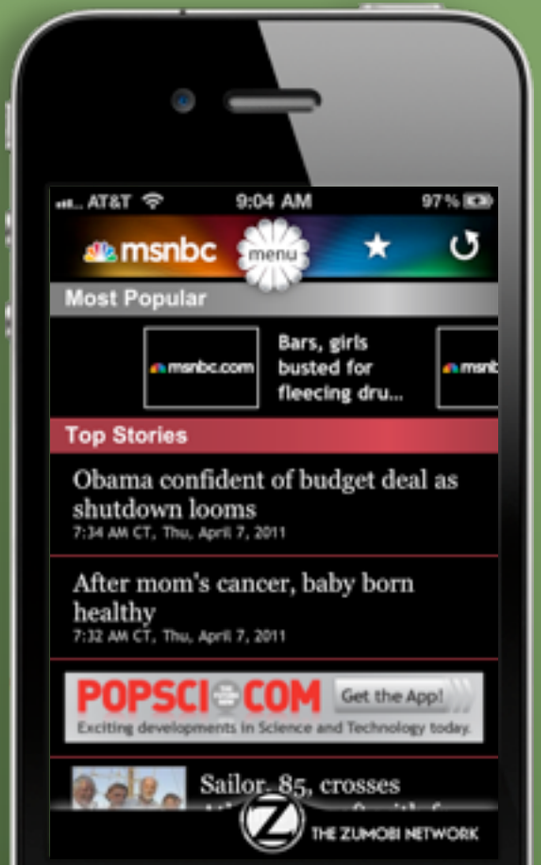

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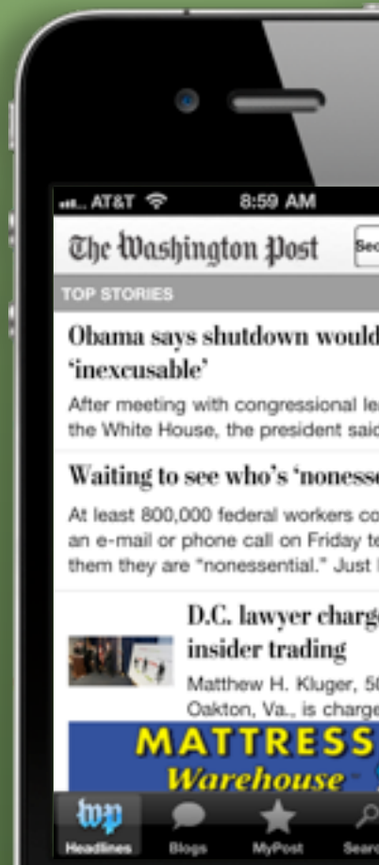

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Togetherness in Libya
Petri-dish economies
Everybody needs good neighbours
Military strategy
Cry havoc! And let slip the math...
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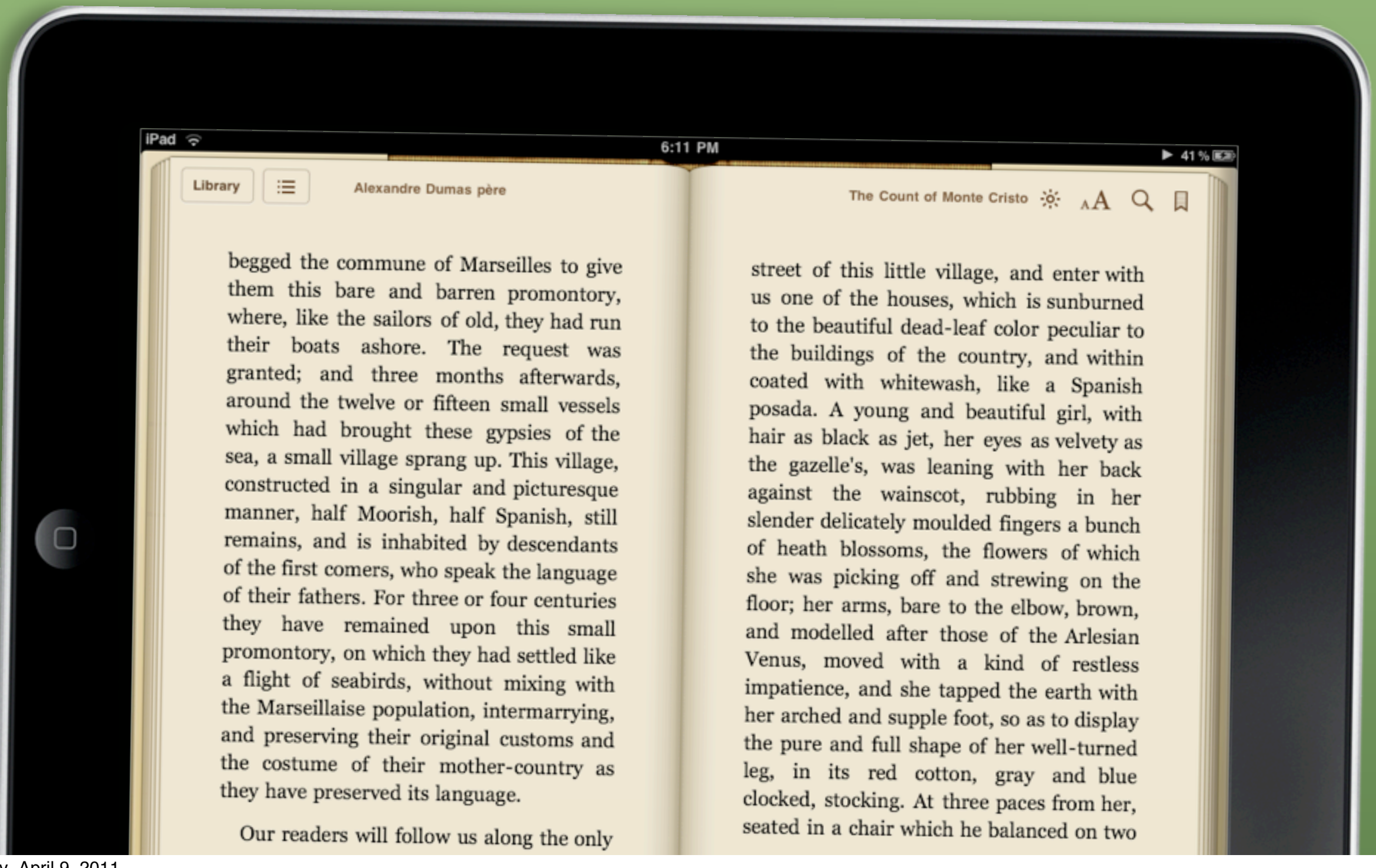

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Top Stories
Obama confident of budget deal as shutdown looms
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After mom's cancer, baby born healthy
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Obama says shutdown would
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D.C. lawyer charg
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Augmented Reality



Immersive



Saturday, April 9, 2011

Location


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10:30 PM
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Chicago
Latest news
Filter
700-719 W. Chicago Ave.
Wednesday, July 21, 2010
Locations in the media
Chicago Real Estate Daily
Trading firm signs big expansion...
Trading firm signs big expansion at 600 W. Chicago
600 W. Chicago
Real estate listings
700 N Larrabee St #1113 listed on...
700 N Larrabee St #1113 listed on Trulia for $399,900, July 21
700 N Larrabee St
Crimes
Narcotics: Manufacturing/delivery...
Sidewalk. Reported 10:54 a.m. July 12.
AT&T
10:38 PM
92 %
Cities
Wrigleyville
Lincoln Park
Roscoe Village
Bucktown
Wicker Park
West Town
Chicago metropolitan area
Goose Island
Cabrini-Green
Near West Side
Chicago Harbor
Grant Park
Dwight D. Eisenhower Expy
Find cars by
Reserve
Drive
My Zipcar
AT&T
10:35 PM
93 %
foursquare
Jeremy Gilbert
MAYORSHIPS
0
BADGES
6
History
Friends
One River Place
Tue Jul 6
Hotel Monteleone
Fri Jul 2
Vaughan's Lounge
Thu Jul 1
Bacchanal Wine Bar
Thu Jul 1
Hotel Monteleone
Thu Jul 1
Friends
Places
Tips
Jeremy

Gyroscope


QUAD LASER
SCORE:
00002000
100%
x1
The first Star Wars game with augmented reality!

Time Sensitive



Connected


AT&T
11:11 PM
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Distance
Pace
8'05" FASTEST
SLOWEST 18'45"
Google
AT&T
9:59 PM
95%
By Title
By Domain
amazon.com
Amazon: JDG
amazon.com
American Airlines: JDG
aa.com
American Airlines: JPG
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americantaxi.com
Apple iOS Developer: Medill
apple.com
Apple: JDG
apple.com
Appmakr: JDG
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Logins
Accounts
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Notes
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Charge Successful
Intuit GoPayment
Map
Locations
Categories
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Specificity



Knows You


AT&T
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bump
7 people you may know use Bump
Contacts
Photos
Apps
Music
Calendar
Social netwo
AT&T
11:18 PM
100%
Menu
Voicemail
Edit
+13124041997
2:34 PM
Hi. This is Danny called The Chicago Tribune wondered if you were at a staff
+18474671844
12:30 PM
Jeremy, It's, Bob companies in the university library, I. T, 7847 it. No for,
Jessica Parker Gilbert
9:59 AM
Transcript not available
+31242140009
4/4/11
Yeah, Hi Jeremy, It's, Greg Crouch calling. Perhaps

Portable


11:27 PM
86 %
epicurious
Enter Keywords
Search
Favorites
er Desserts
er Dinners
Course Salads
g Entrées
ng/Shower
am
ning Cocktails
t
an Classics
ght Dinners
the cook's companion
AT&T
11:02 PM
epicurio
the cook's companio
AT&T
11:06 PM
Lists
Add a new list
Inbox
Personal
11SP-Usability Module
SND
AEJMC
Groceries
Notification

Things to Avoid

- Unnecessary Fidelity
- Reinvent UI Cues
- Making Affordances Hard to Find/Understand
- Inconsistency
- Bad Error Messages or No User Feedback
- Asking Users to Tell You Things You Know

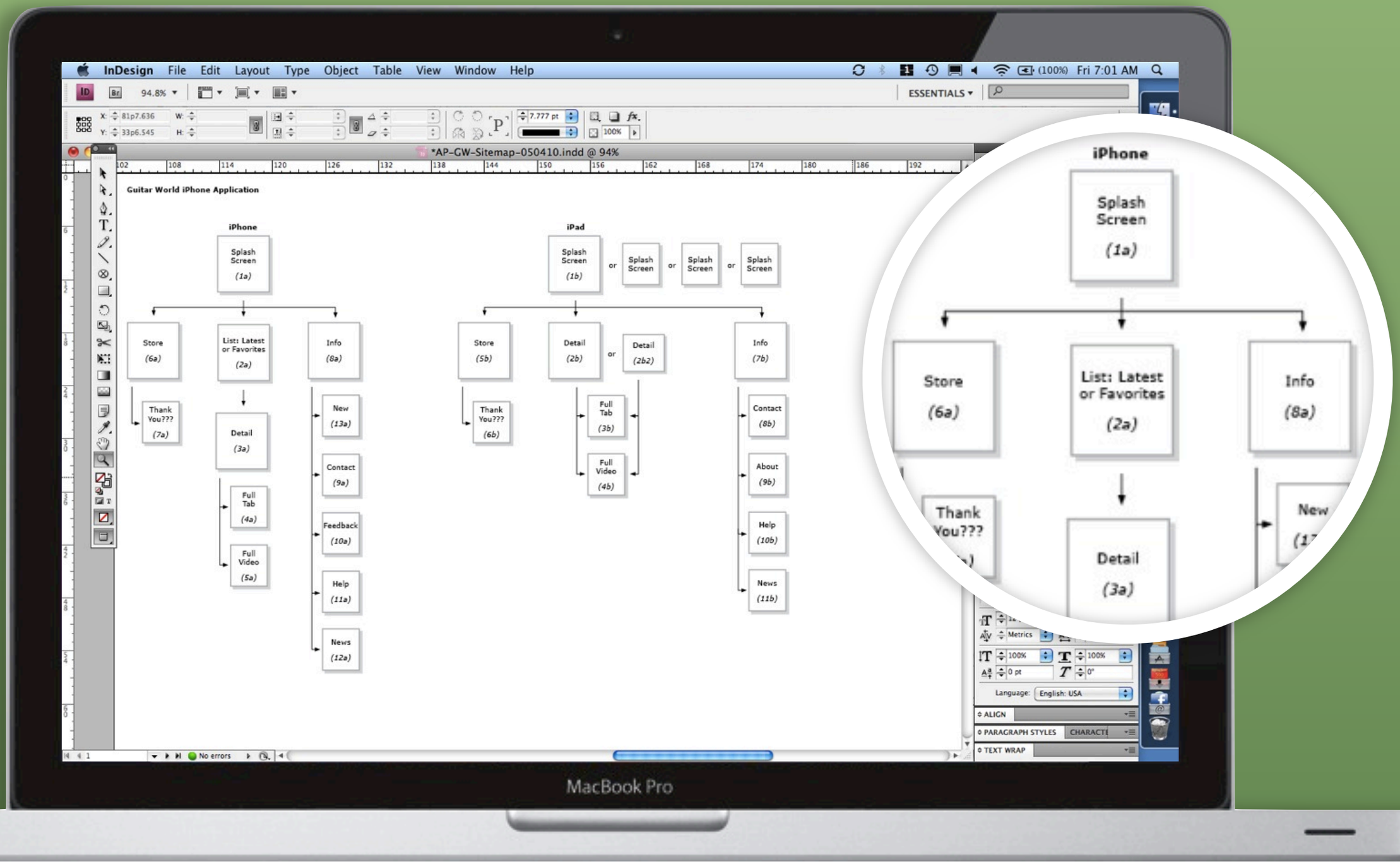
Things to Avoid

- Unnecessary Fidelity
- Reinvent UI Cues
- Making Affordances Hard to Find/Understand
- Inconsistency
- Bad Error Messages or No User Feedback
- Asking Users to Tell You Things You Know

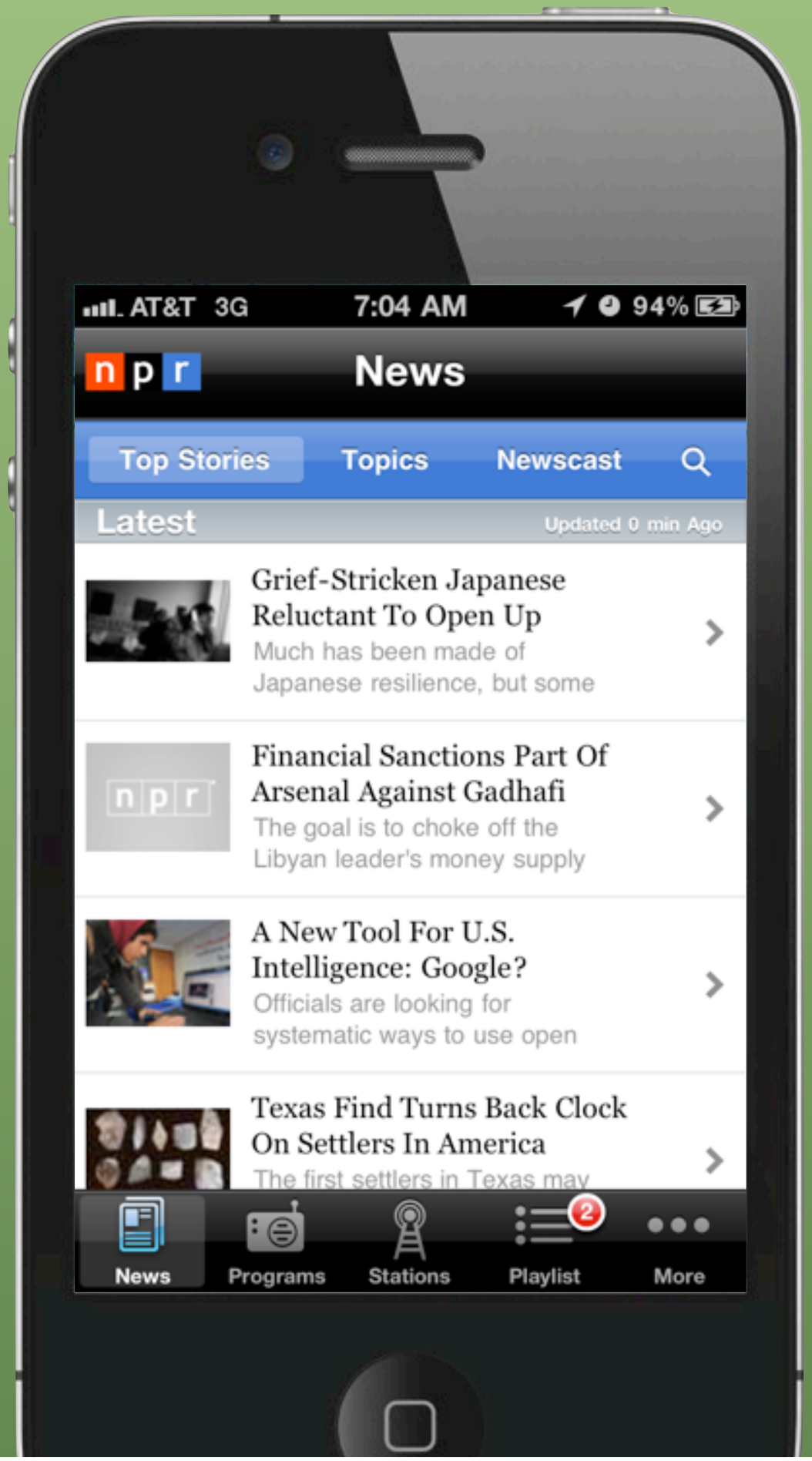


Information Architecture & Prototyping

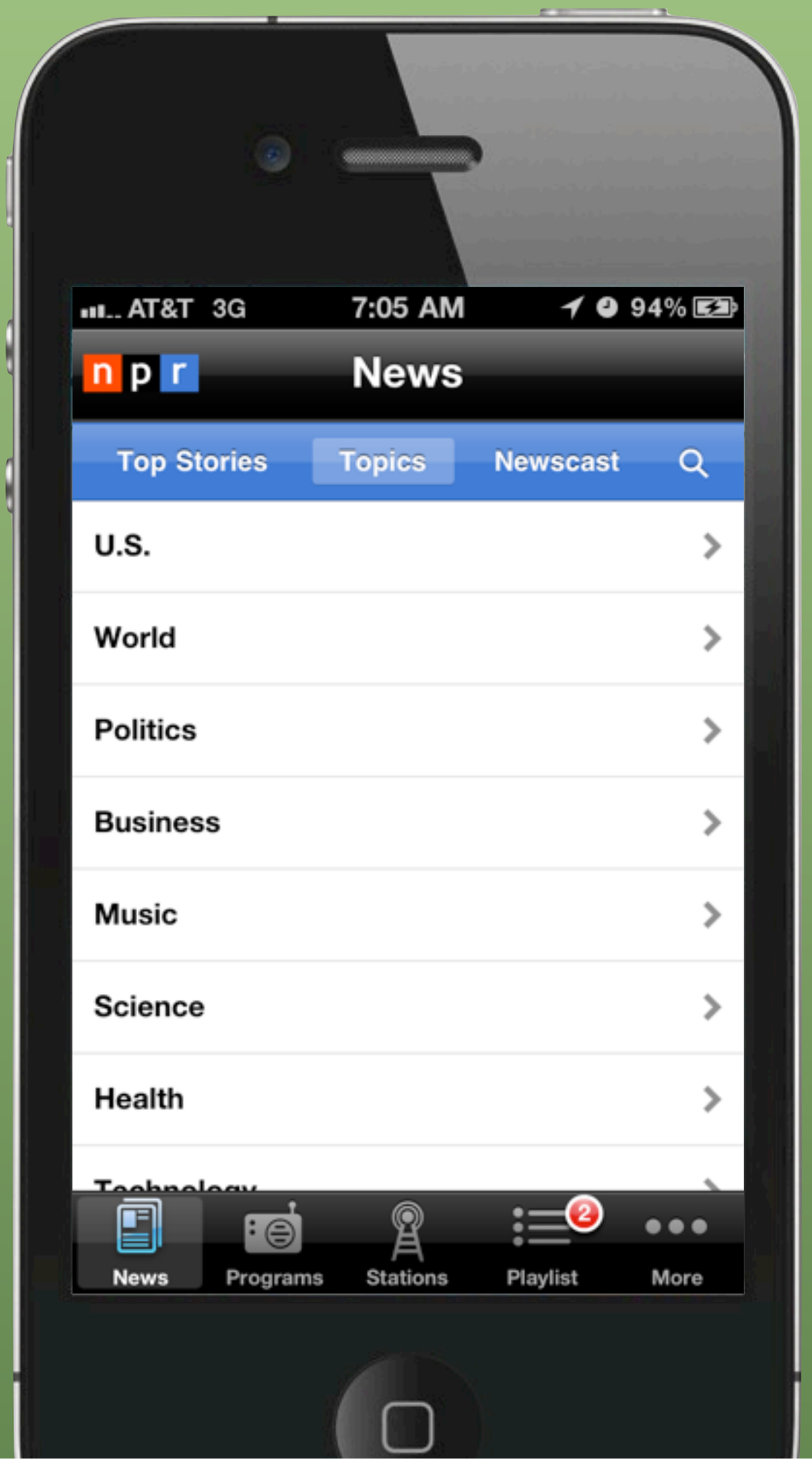
Sitemaps

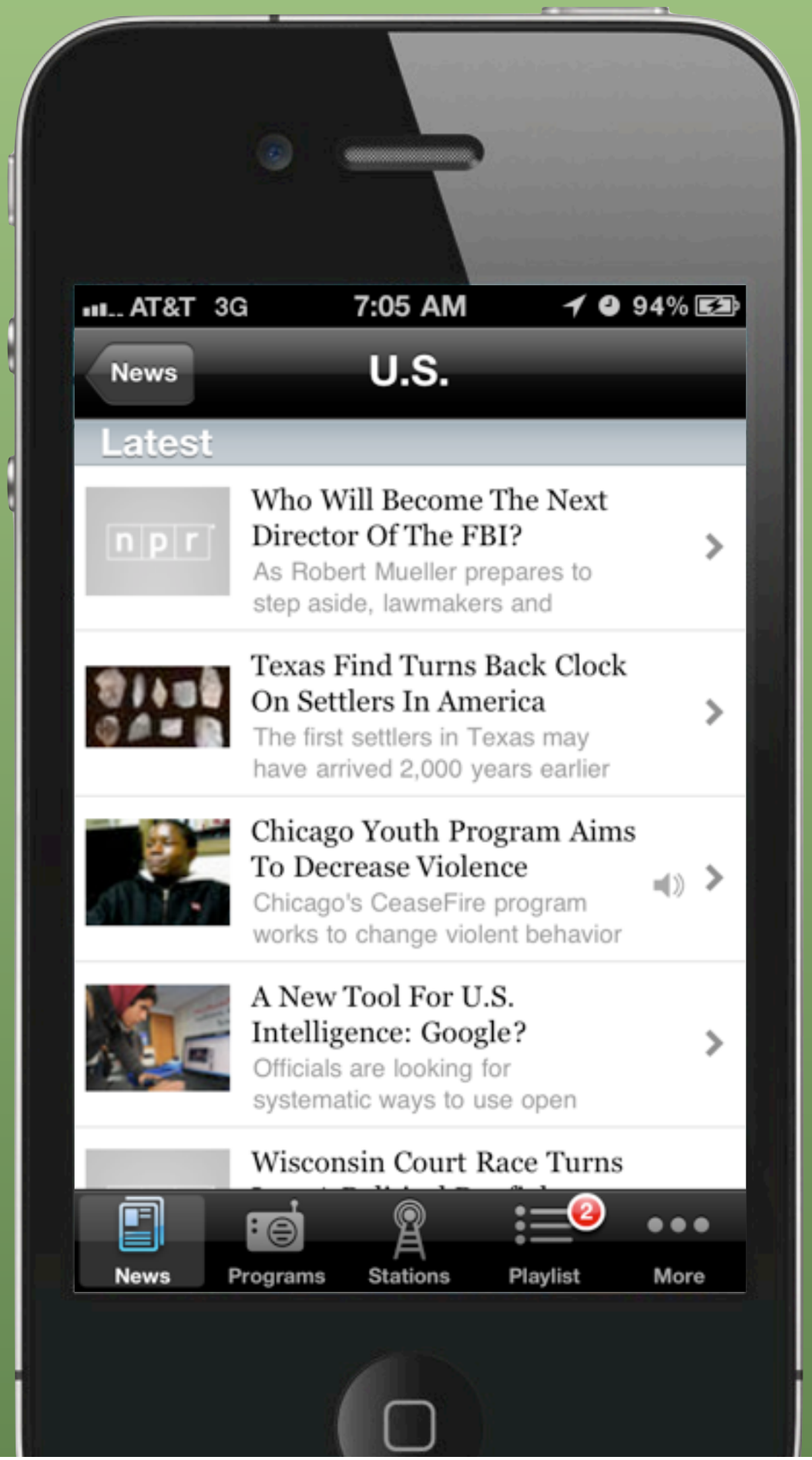

InDesign File Edit Layout Type Object Table View Window Help
*AP-GW-Sitemap-050410.indd @ 94%
Guitar World iPhone Application
iPhone
Splash Screen (1a)
Store (6a)
List: Latest or Favorites (2a)
Info (8a)
Thank You??? (7a)
Detail (3a)
Full Tab (4a)
Full Video (5a)
New (13a)
Contact (9a)
Feedback (10a)
Help (11a)
News (12a)
iPad
Splash Screen (1b)
or
Splash Screen
Store (5b)
Detail (2b)
Detail (2b2)
Thank You??? (6b)
Full Tab (3b)
Full Video (4b)
Info (7b)
Contact (8b)
About (9b)
Help (10b)
News (11b)
iPhone
Splash Screen (1a)
Store (6a)
List: Latest or Favorites (2a)
Info (8a)
Thank You???
Detail (3a)
New
MacBook Pro

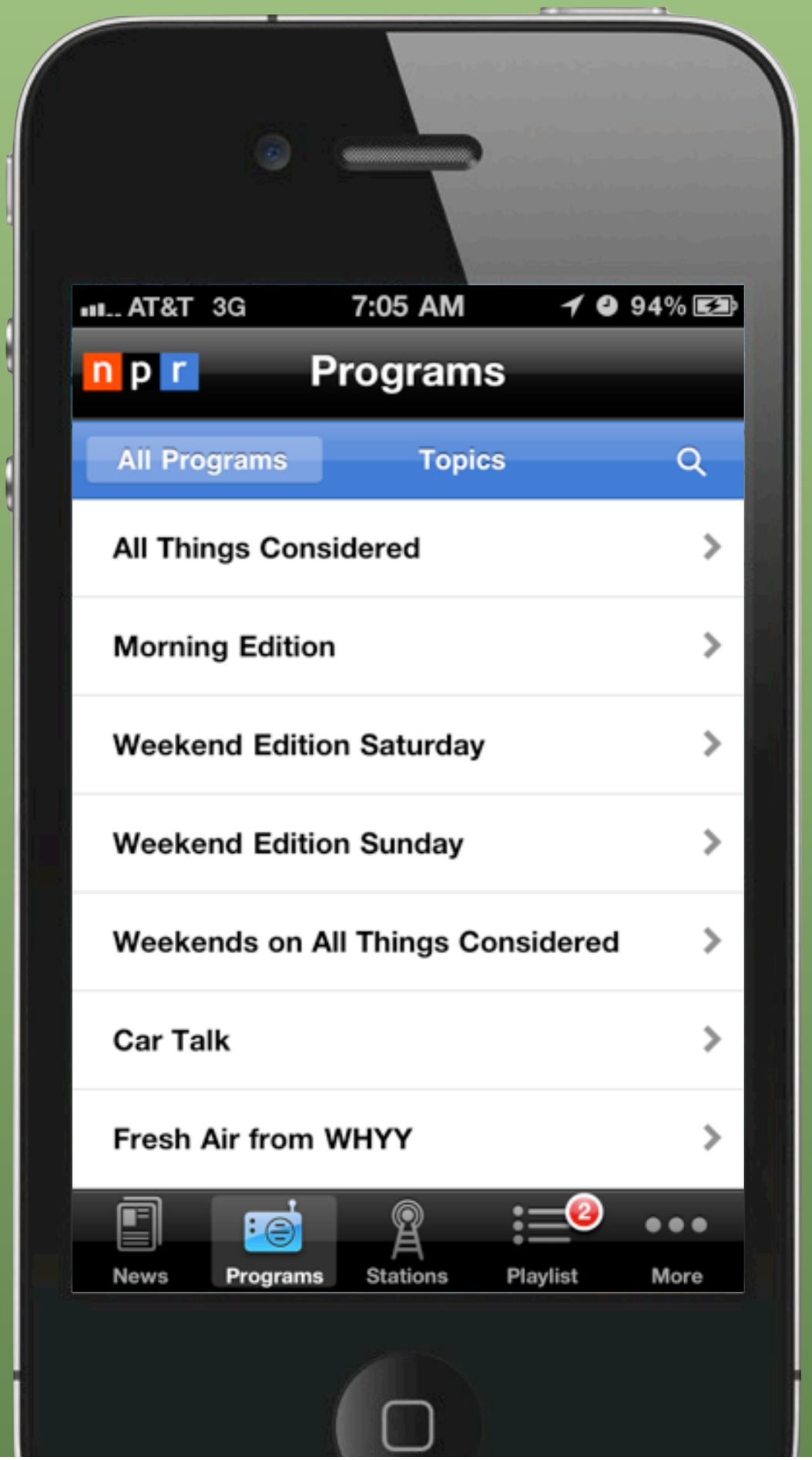

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npr news
DEVELOPED BY bottle rocket

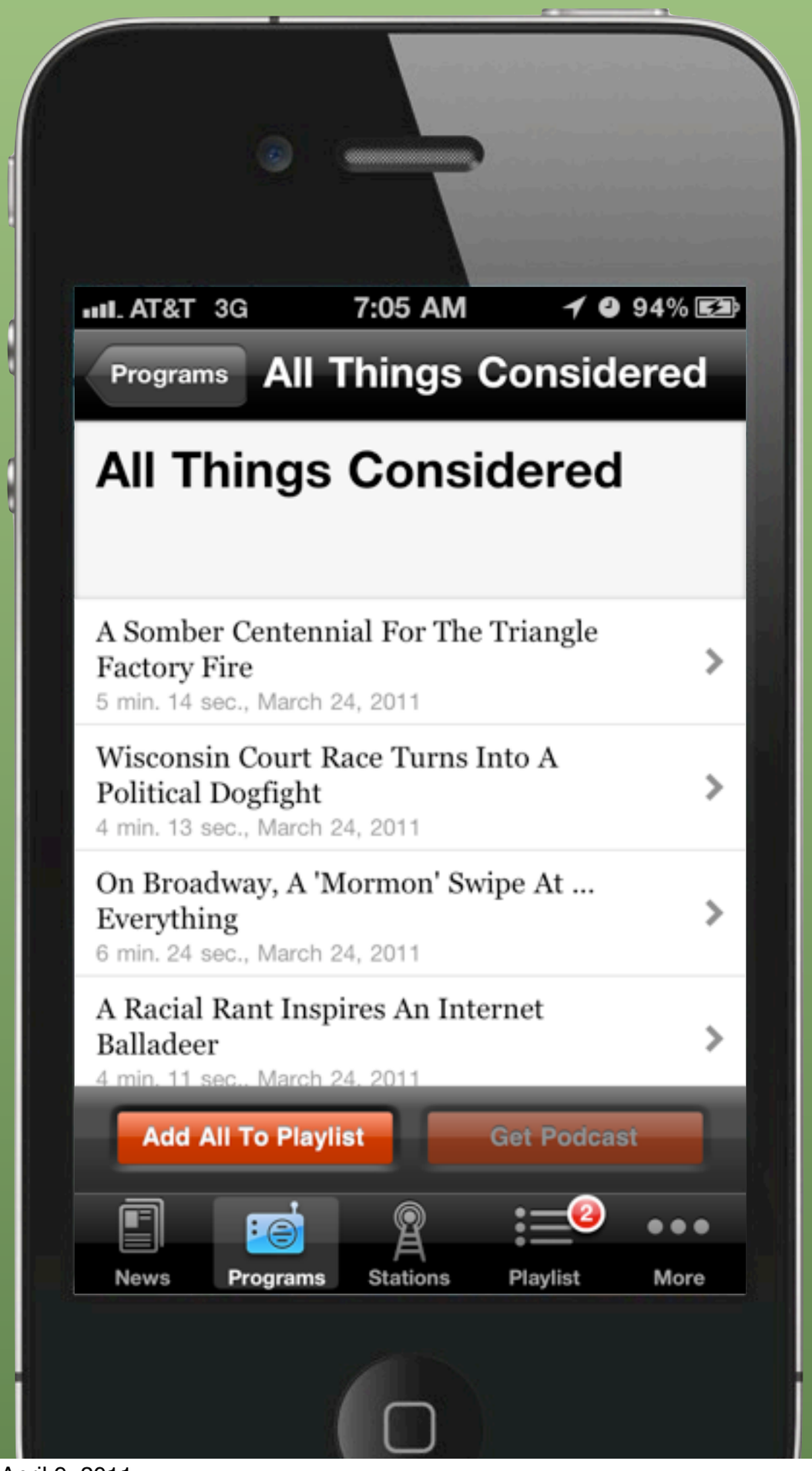

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Grief-Stricken Japanese Reluctant To Open Up
Much has been made of Japanese resilience, but some
Financial Sanctions Part Of Arsenal Against Gadhafi
The goal is to choke off the Libyan leader's money supply
A New Tool For U.S. Intelligence: Google?
Officials are looking for systematic ways to use open
Texas Find Turns Back Clock On Settlers In America
The first settlers in Texas may
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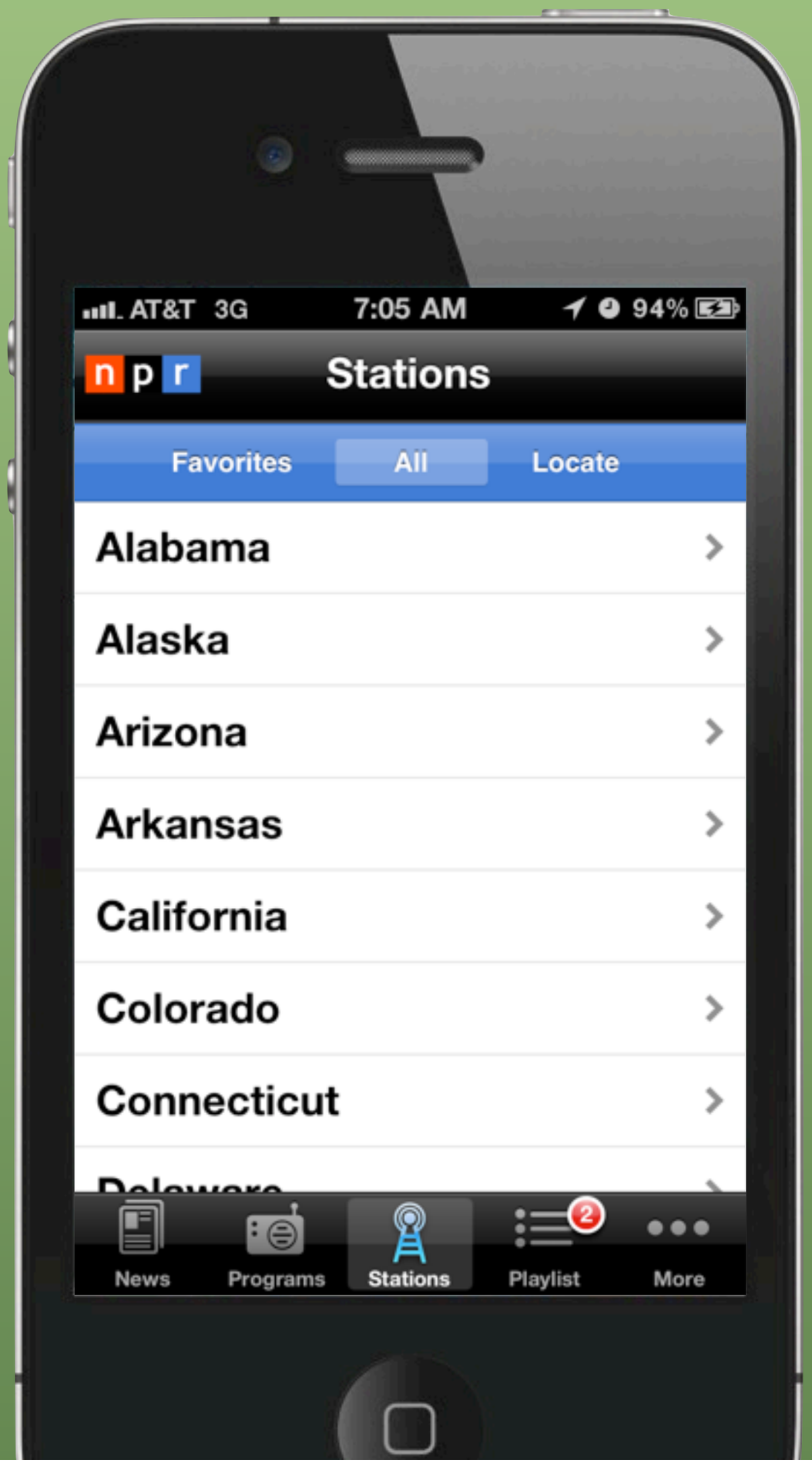

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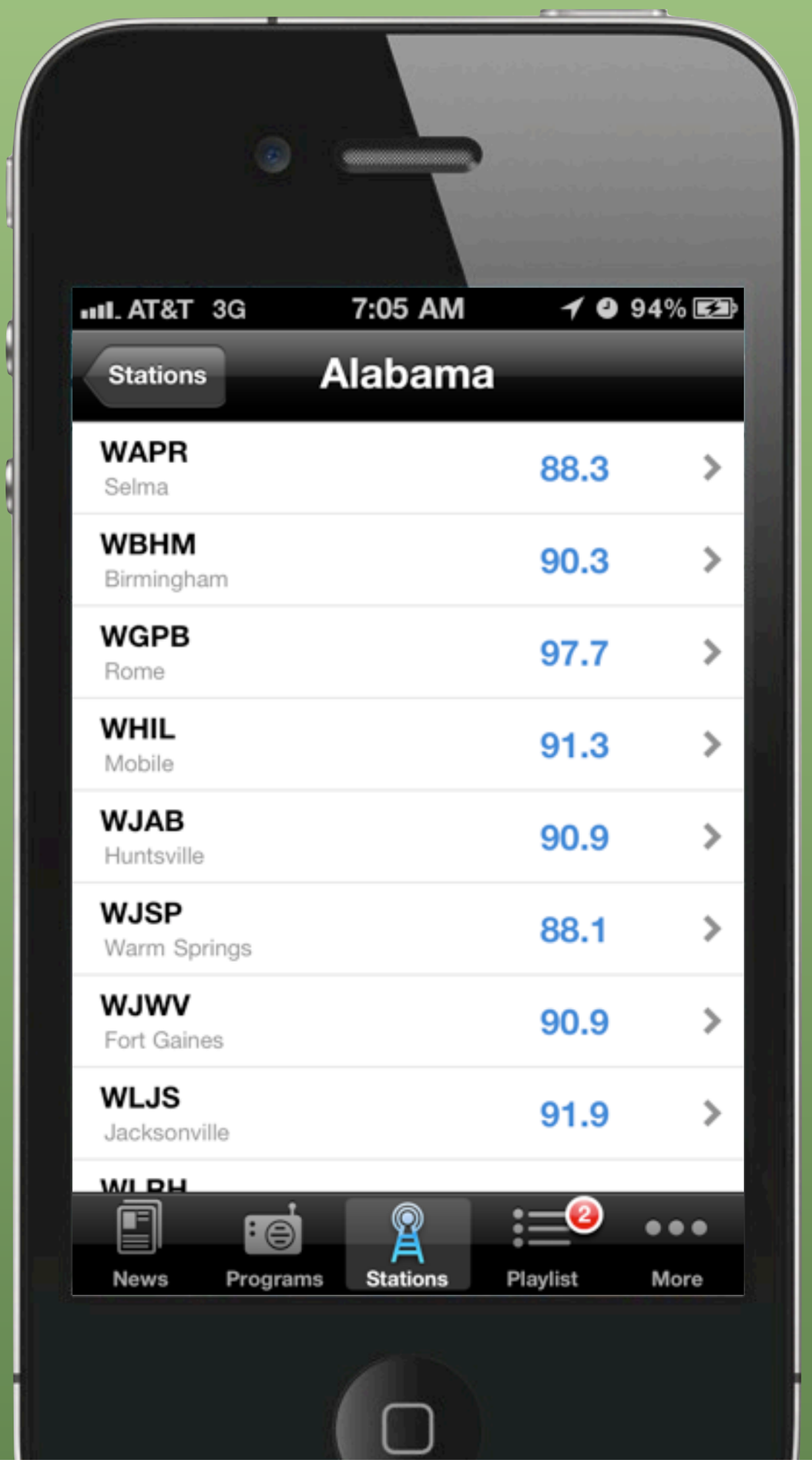

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U.S.
Latest
Who Will Become The Next Director Of The FBI?
As Robert Mueller prepares to step aside, lawmakers and
Texas Find Turns Back Clock On Settlers In America
The first settlers in Texas may have arrived 2,000 years earlier
Chicago Youth Program Aims To Decrease Violence
Chicago's CeaseFire program works to change violent behavior
A New Tool For U.S. Intelligence: Google?
Officials are looking for systematic ways to use open
Wisconsin Court Race Turns
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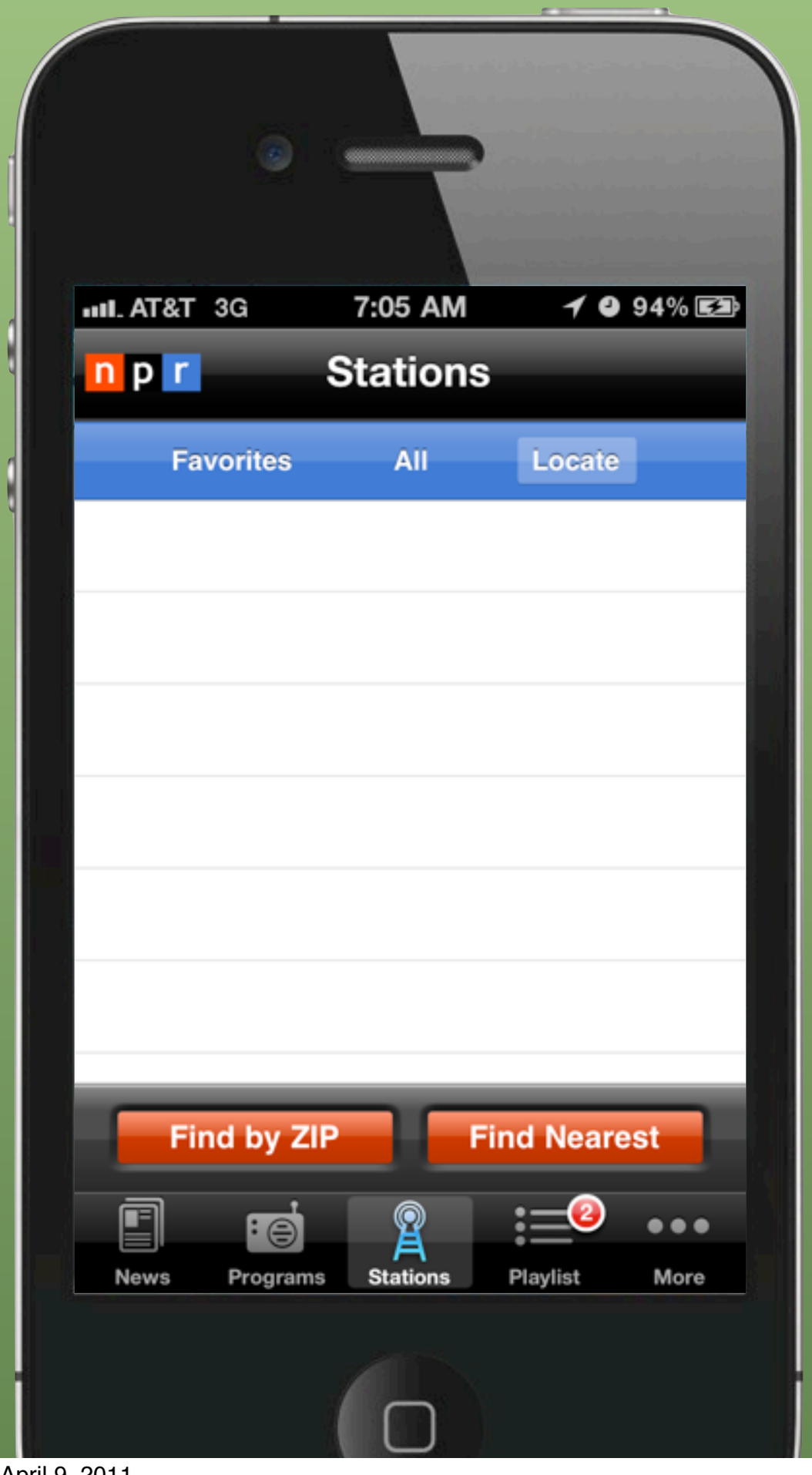

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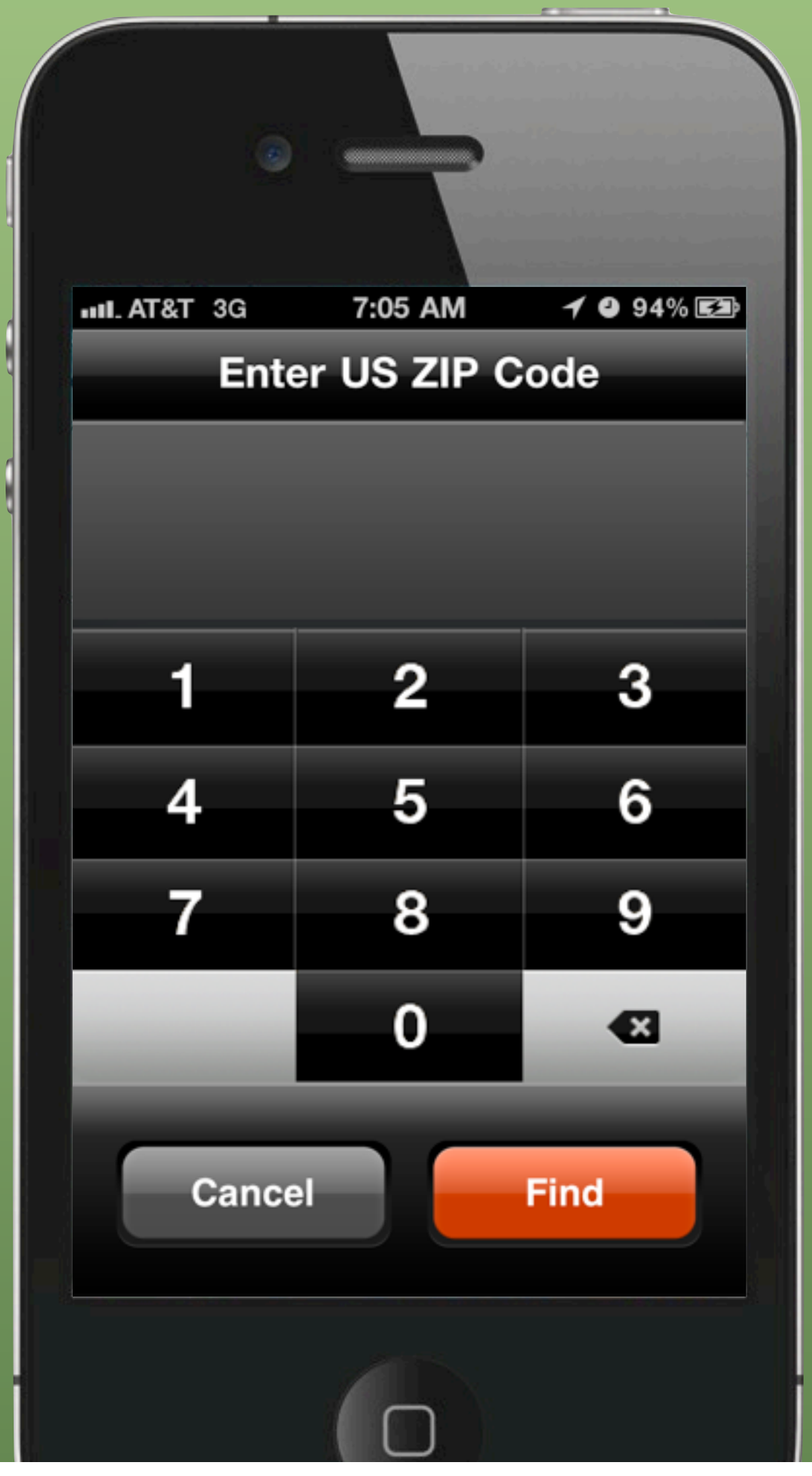

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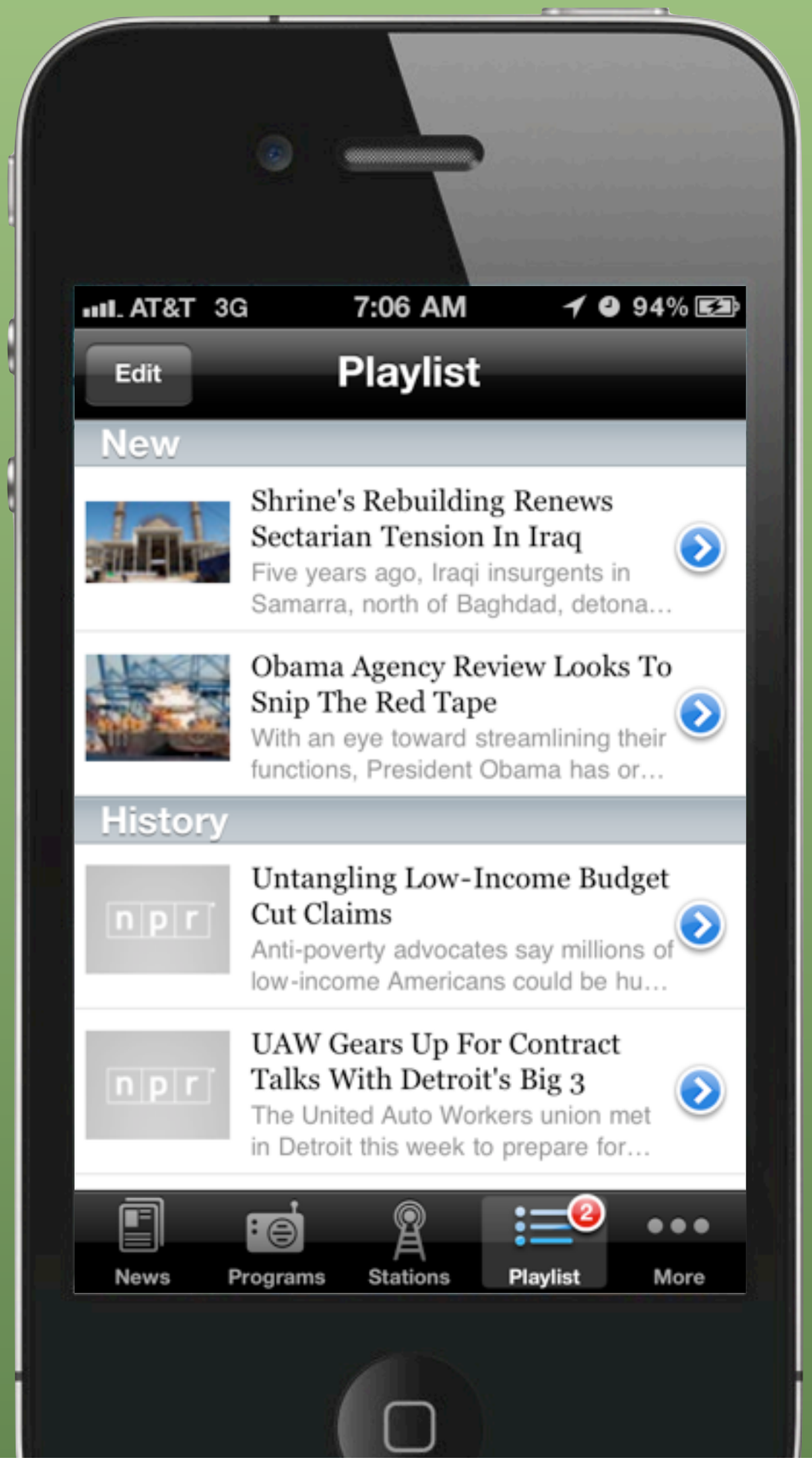

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Who Will Become The Next Director Of The FBI?
by Carrie Johnson - March 25, 2011
Morning Edition
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Over the next several weeks, the Obama administration will have a big decision on its hands: choosing the next director of the FBI. The 10-year term of the FBI's current leader, Robert Mueller, expires in September. A committee is already searching for candidates to replace him.
Mueller became the FBI director in September 2001, days before the terrorist attacks that would change the bureau's
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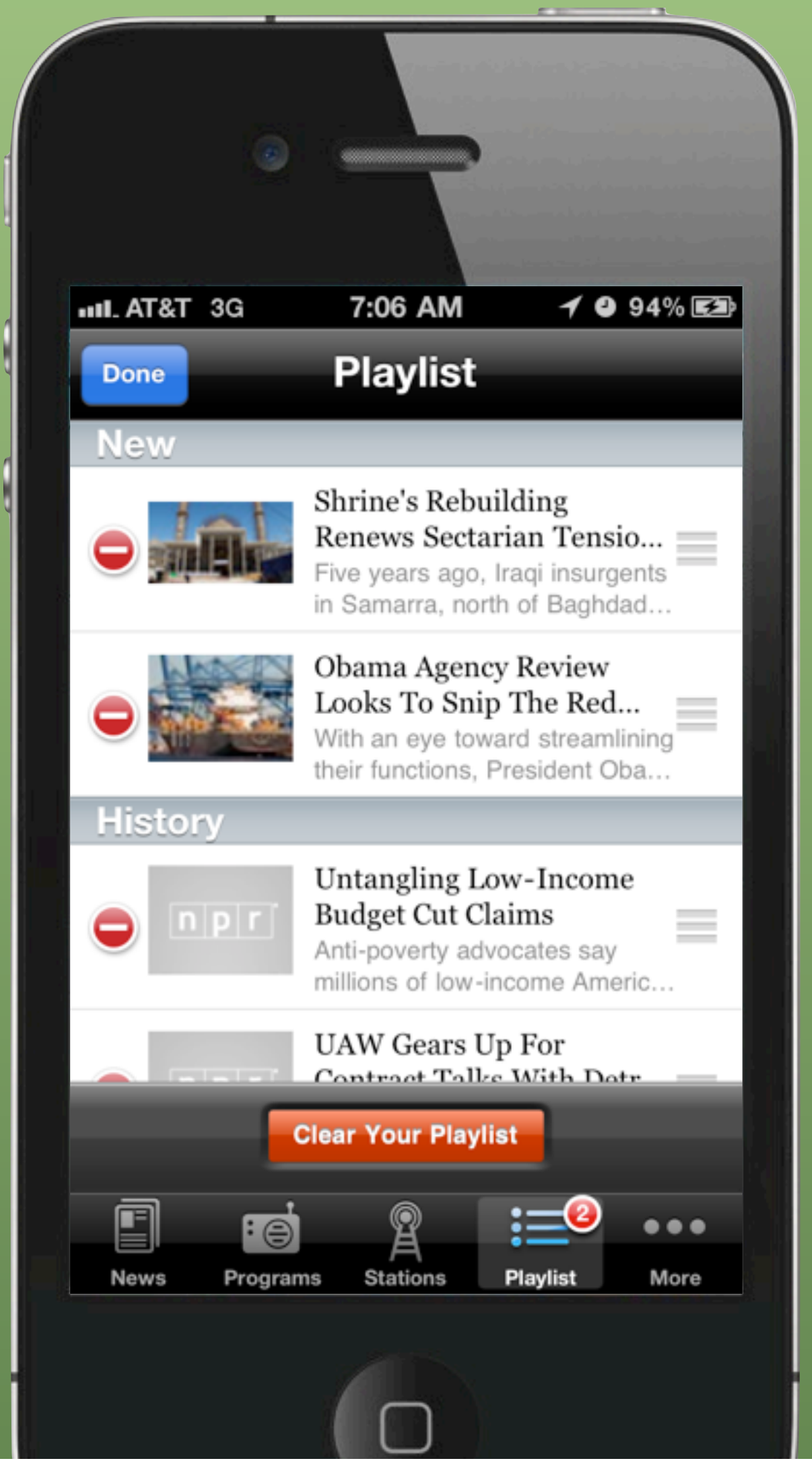

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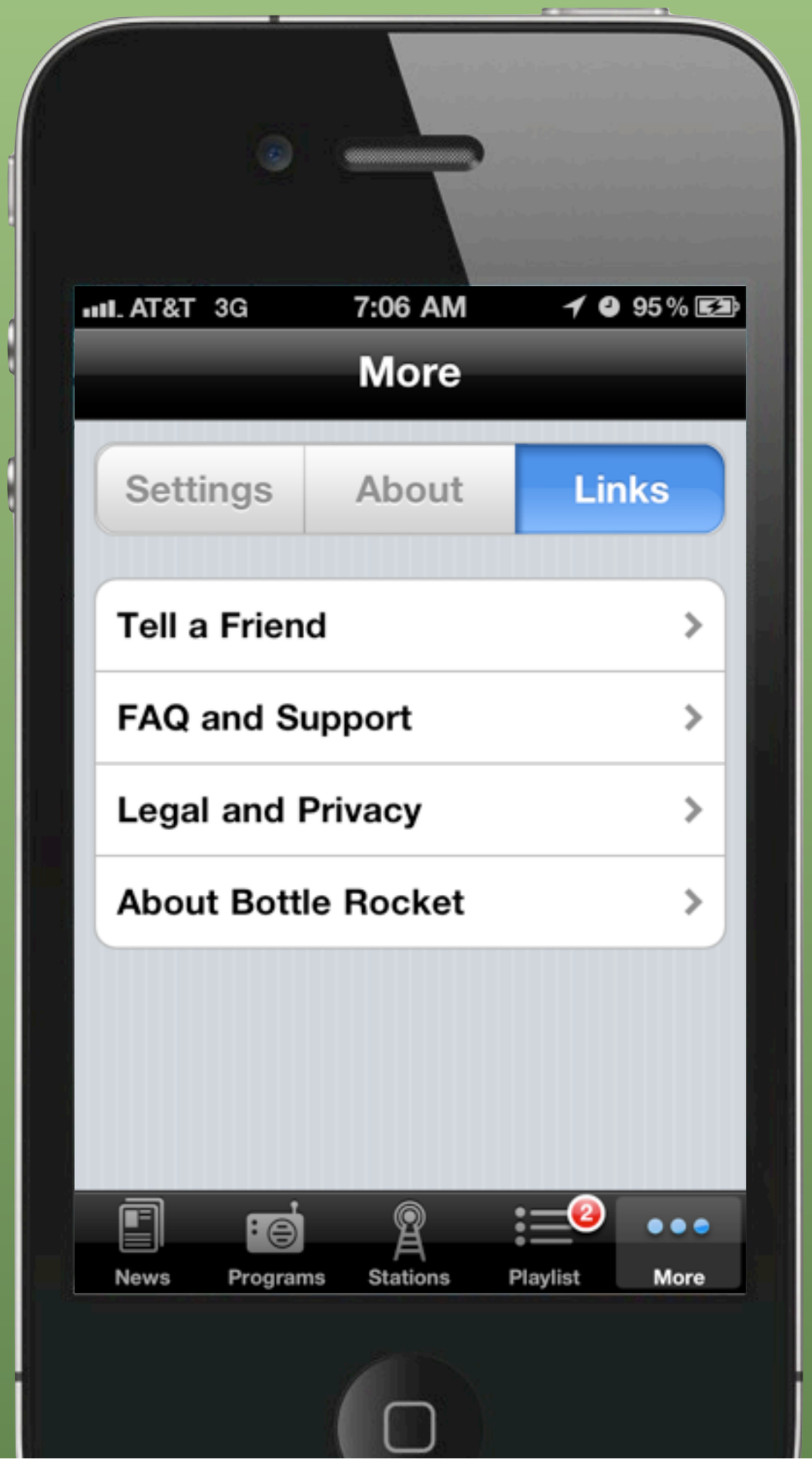

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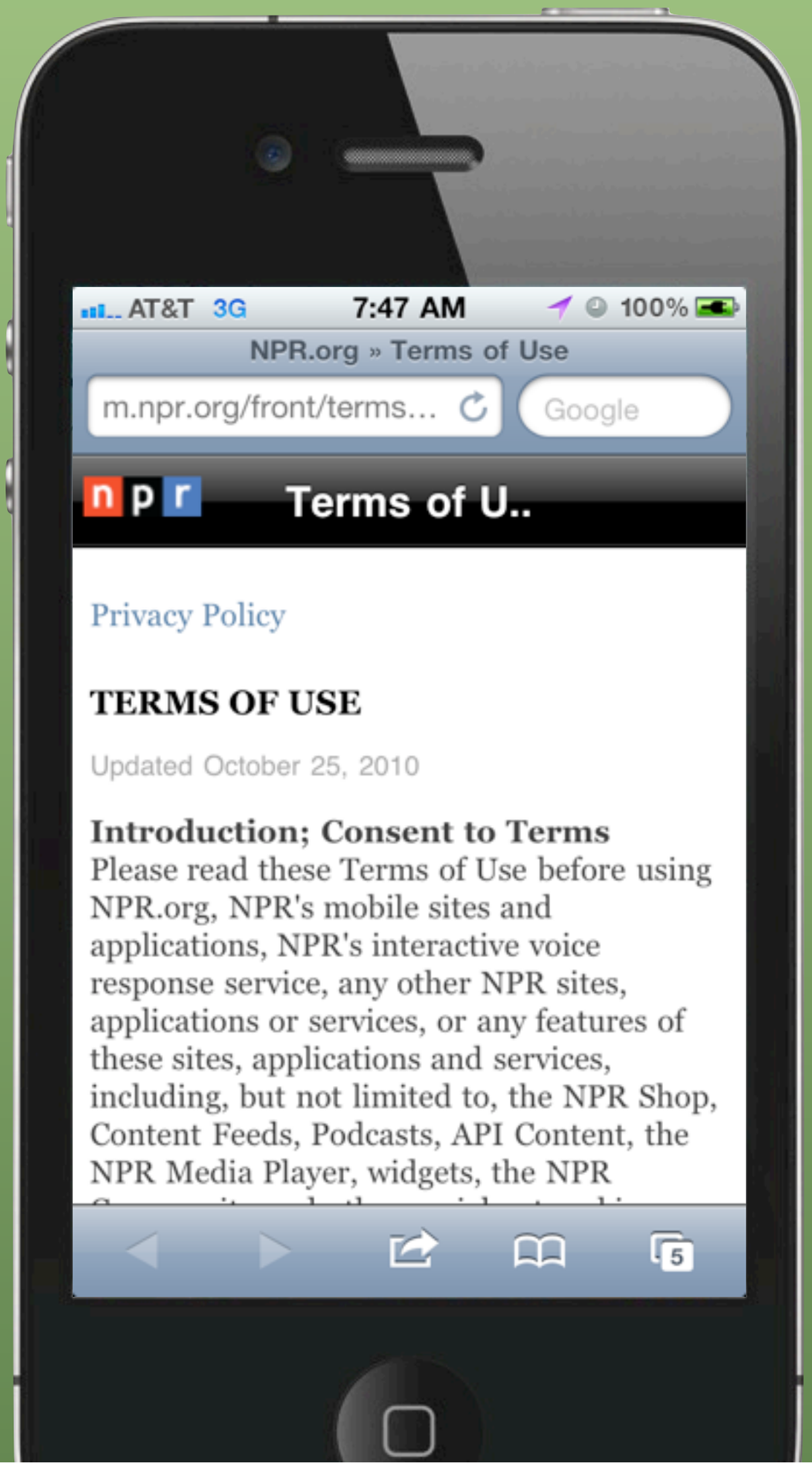

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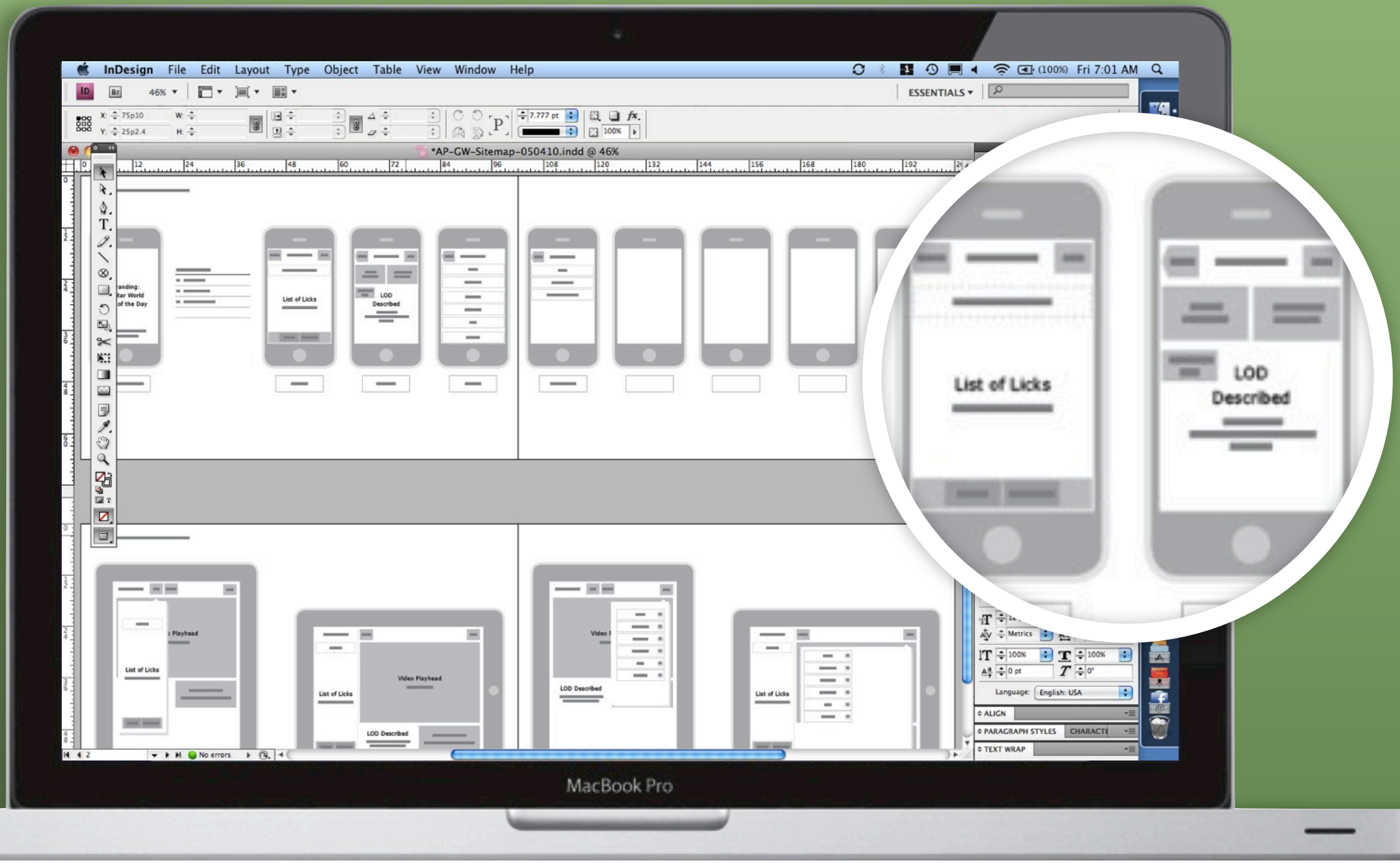

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Wireframing



Wireframing

App	Link: bit.ly/SND-MobileDesign-Wireframe
Photoshop	bit.ly/SND-MobileDesign-Wireframe-Photoshop
Illustrator	bit.ly/SND-MobileDesign-Wireframe-Illustrator
Indesign	yhoo.it/SND-MobileDesign-Wireframe-Indesign
Stencil	bit.ly/SND-MobileDesign-Wireframe-Stencil
Web-Based	bit.ly/SND-MobileDesign-Wireframe-Browser
Keynote	bit.ly/SND-MobileDesign-Wireframe-Keynote
OmniGraffle	yhoo.it/SND-MobileDesign-Wireframe-OmniGraffle

Designing for Small Screens

Gestures, Not Clicks

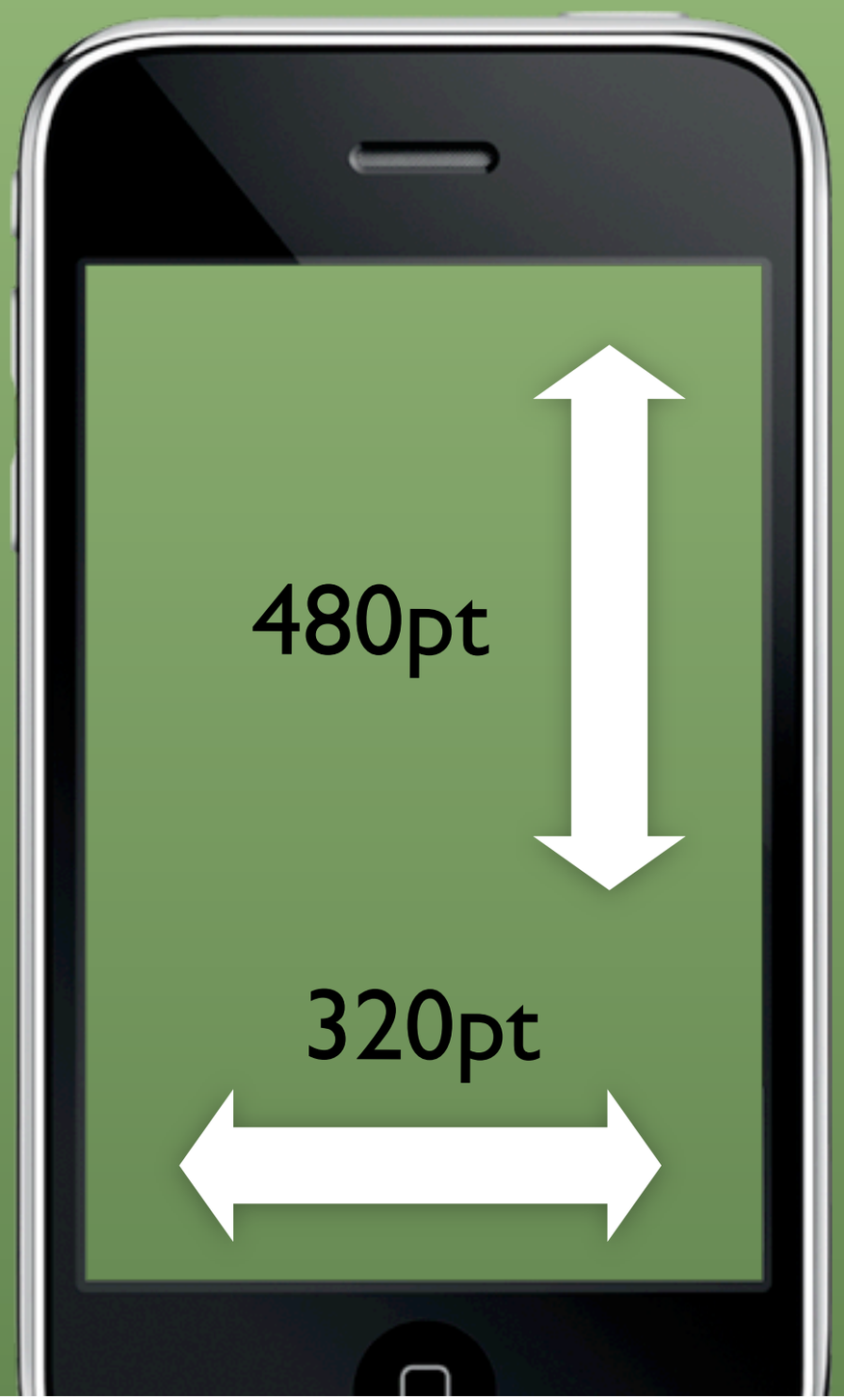
Gesture	Action
Tap	To press or select a control or item
Drag	Scroll or pan
Flick	Scroll or pan quickly
Swipe	In a table-view row, to reveal the Delete button
Double Tap	Zoom in and center or zoom out (if already zoomed in)
Pinch Open	Zoom in
Pinch Close	Zoom out
Touch & Hold	Edit text (select, select all, copy)
Shake	Undo

Apple Human Interface Guidelines

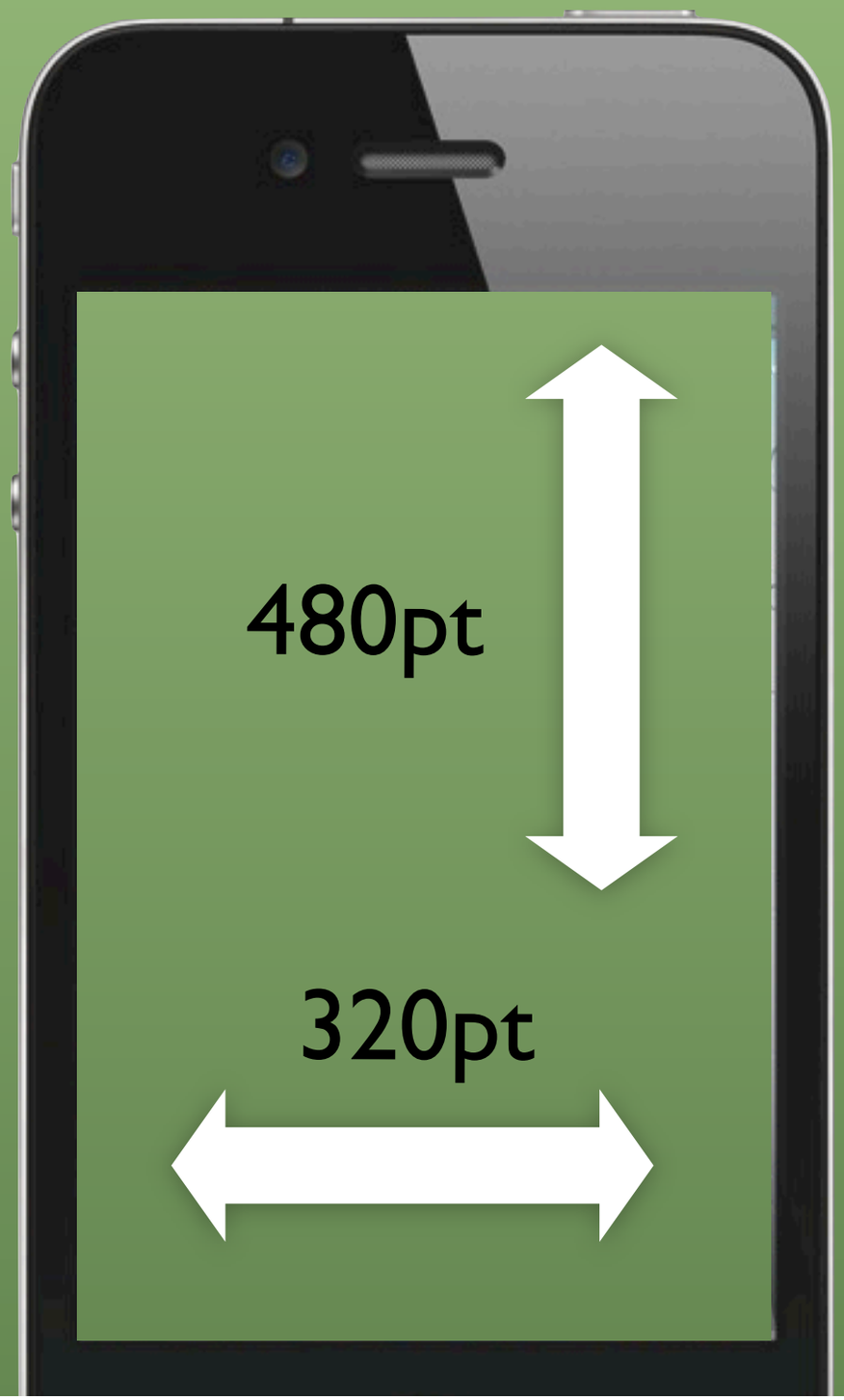
Points vs Pixels

- **Pixel** describes the size of a device screen or the size of an icon you create in an image-editing application.
- **Point** describes the size of an area that is drawn onscreen.

iPhone 3G



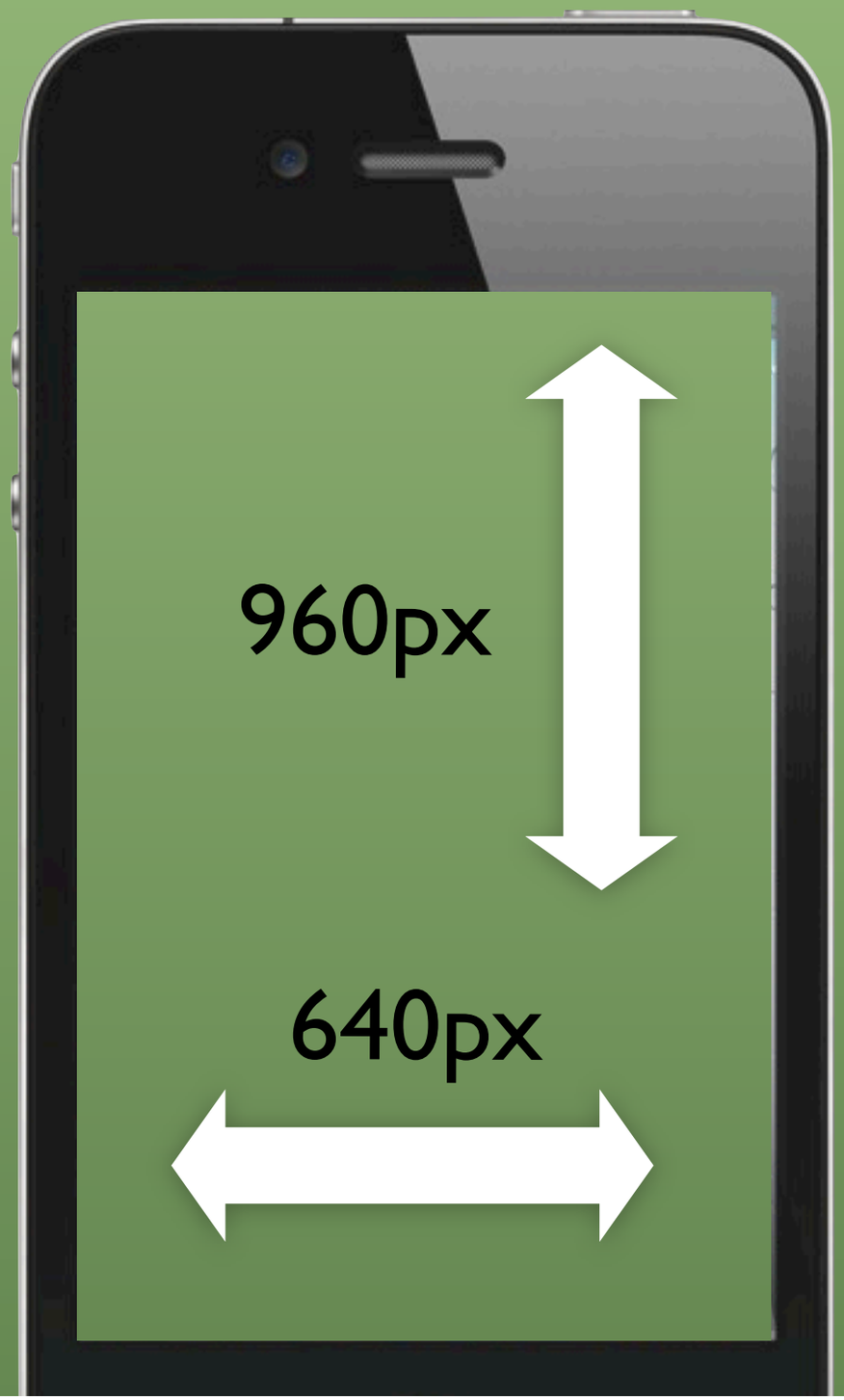
iPhone 4

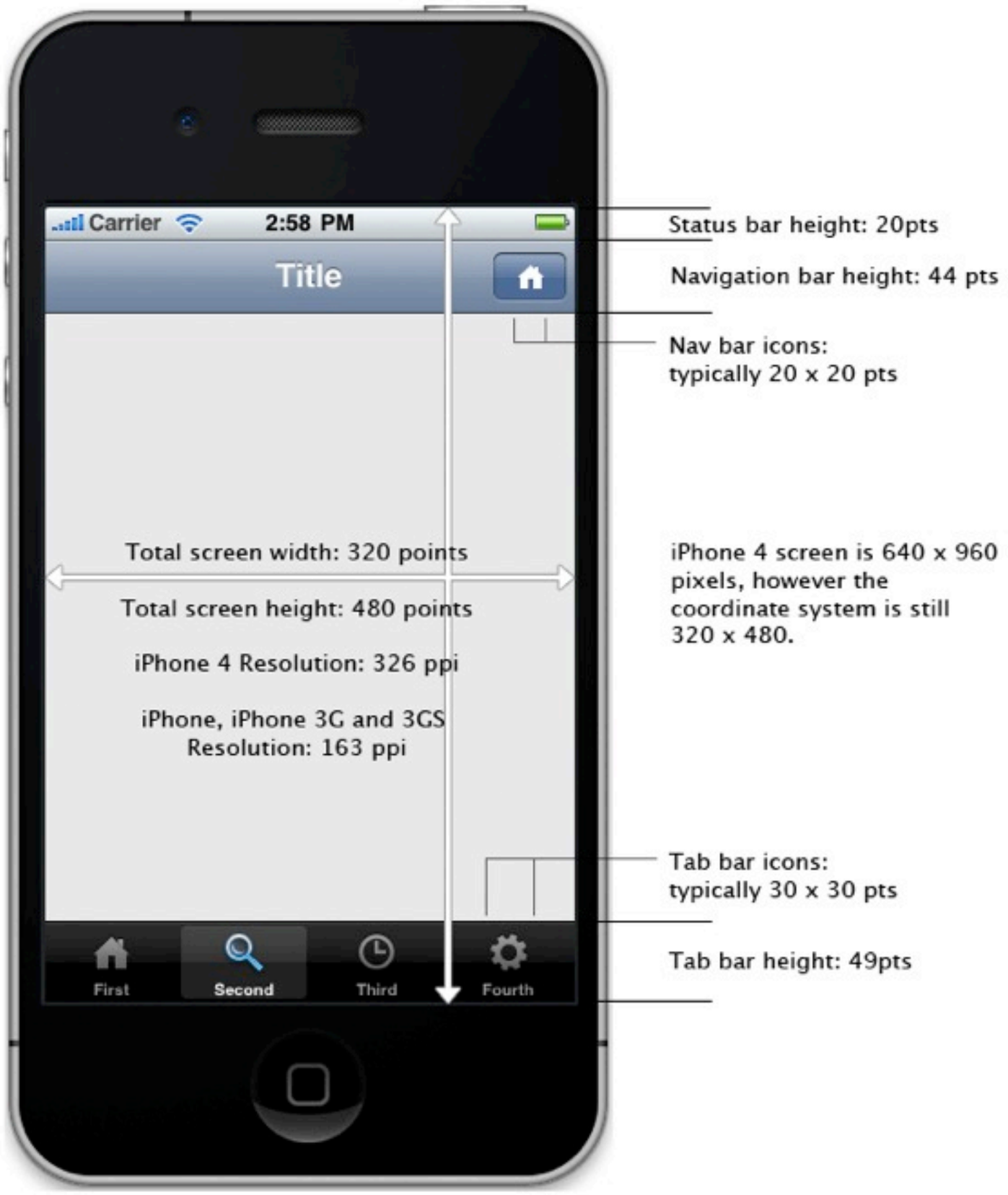


iPhone 3G



iPhone 4




Carrier
2:58 PM
Title
Status bar height: 20pts
Navigation bar height: 44 pts
Nav bar icons:
typically 20 x 20 pts
Total screen width: 320 points
Total screen height: 480 points
iPhone 4 Resolution: 326 ppi
iPhone, iPhone 3G and 3GS
Resolution: 163 ppi
iPhone 4 screen is 640 x 960
pixels, however the
coordinate system is still
320 x 480.
First
Second
Third
Fourth
Tab bar icons:
typically 30 x 30 pts
Tab bar height: 49pts

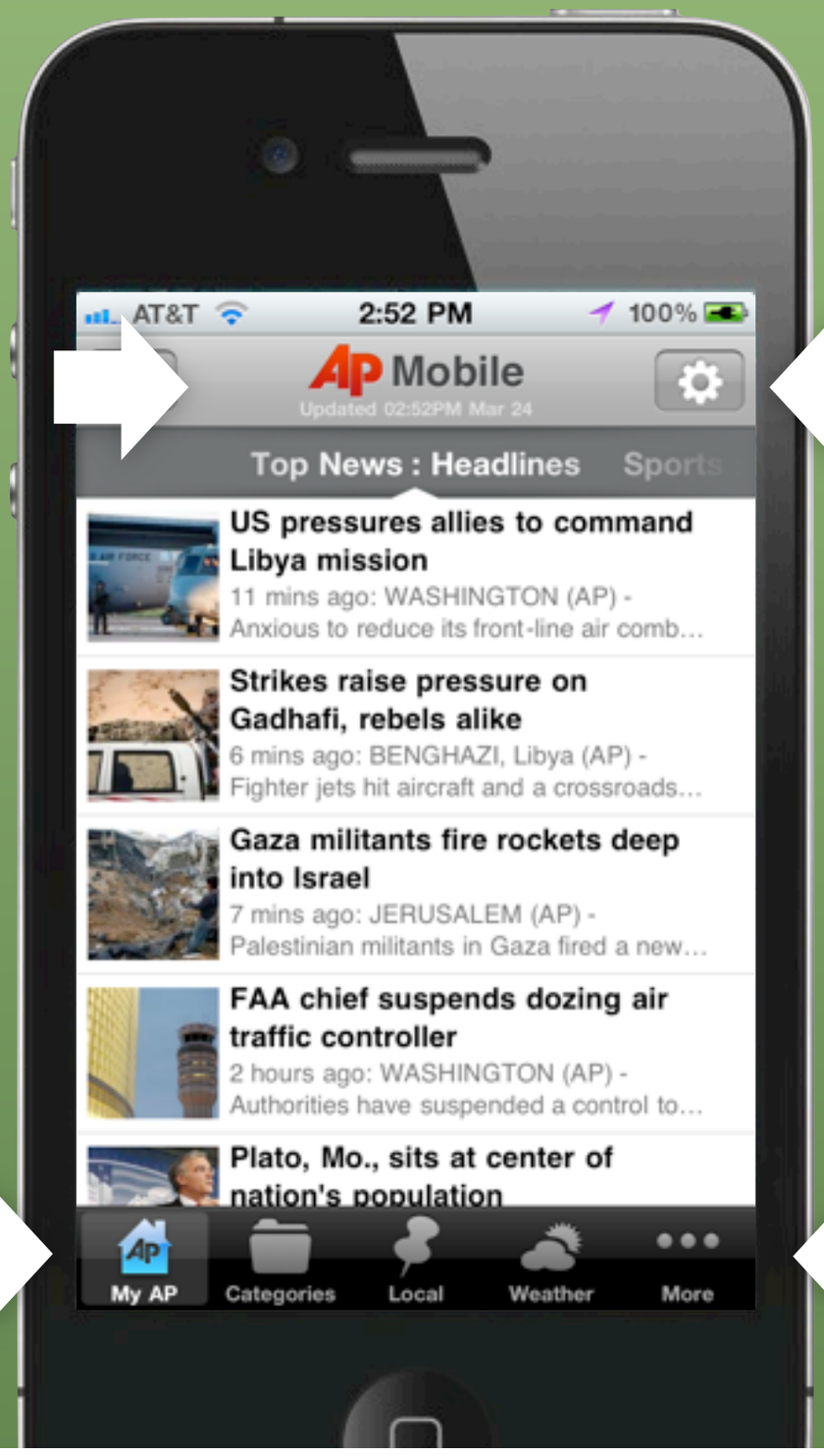
Finger-Sized

44pts



UI Conventions

Location



Settings

Primary Nav

Additional Nav

Multitasking



Apple Human Interface Guidelines

~~Multitasking~~

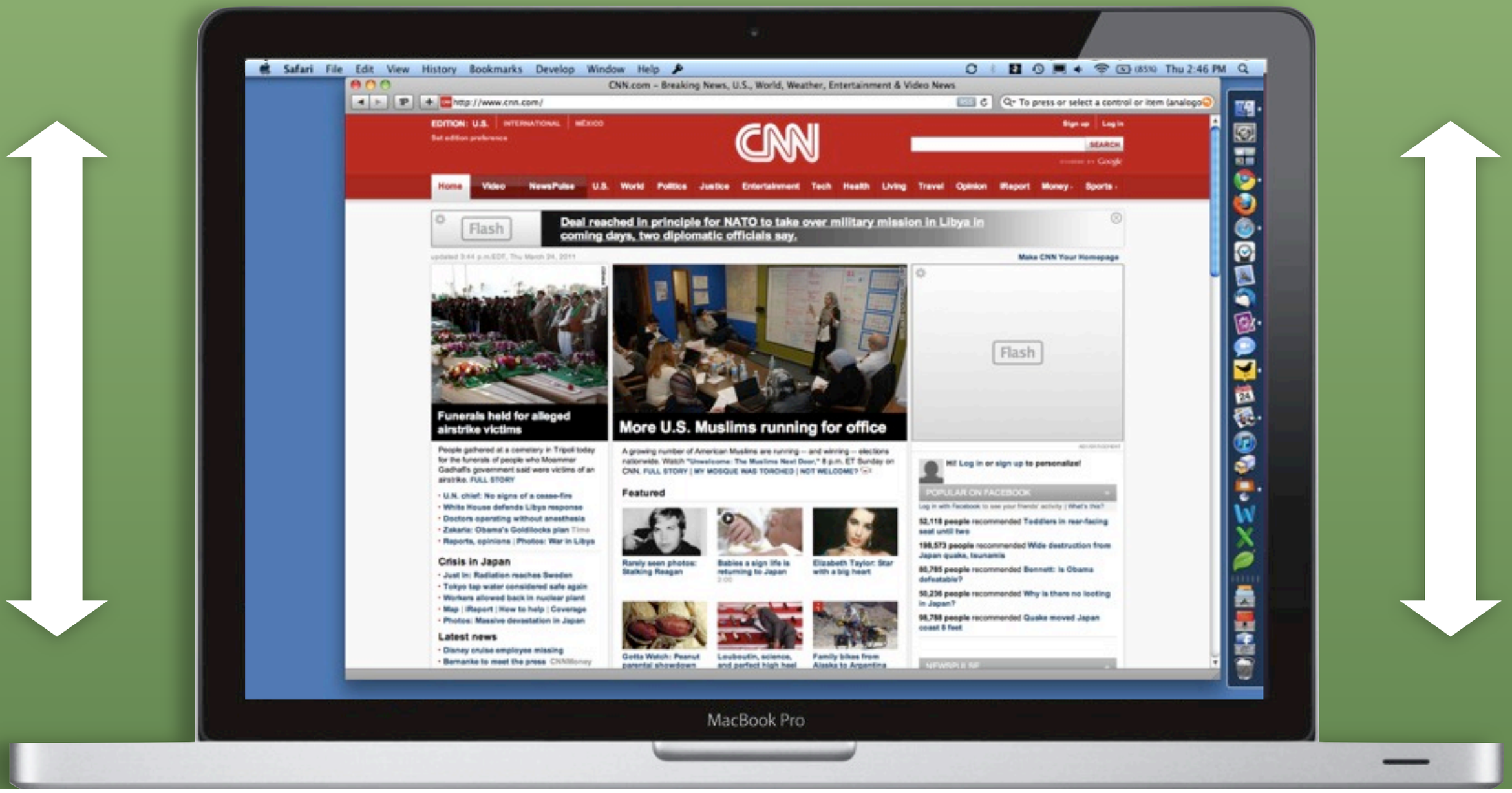


Apple Human Interface Guidelines



Cards vs Scroll

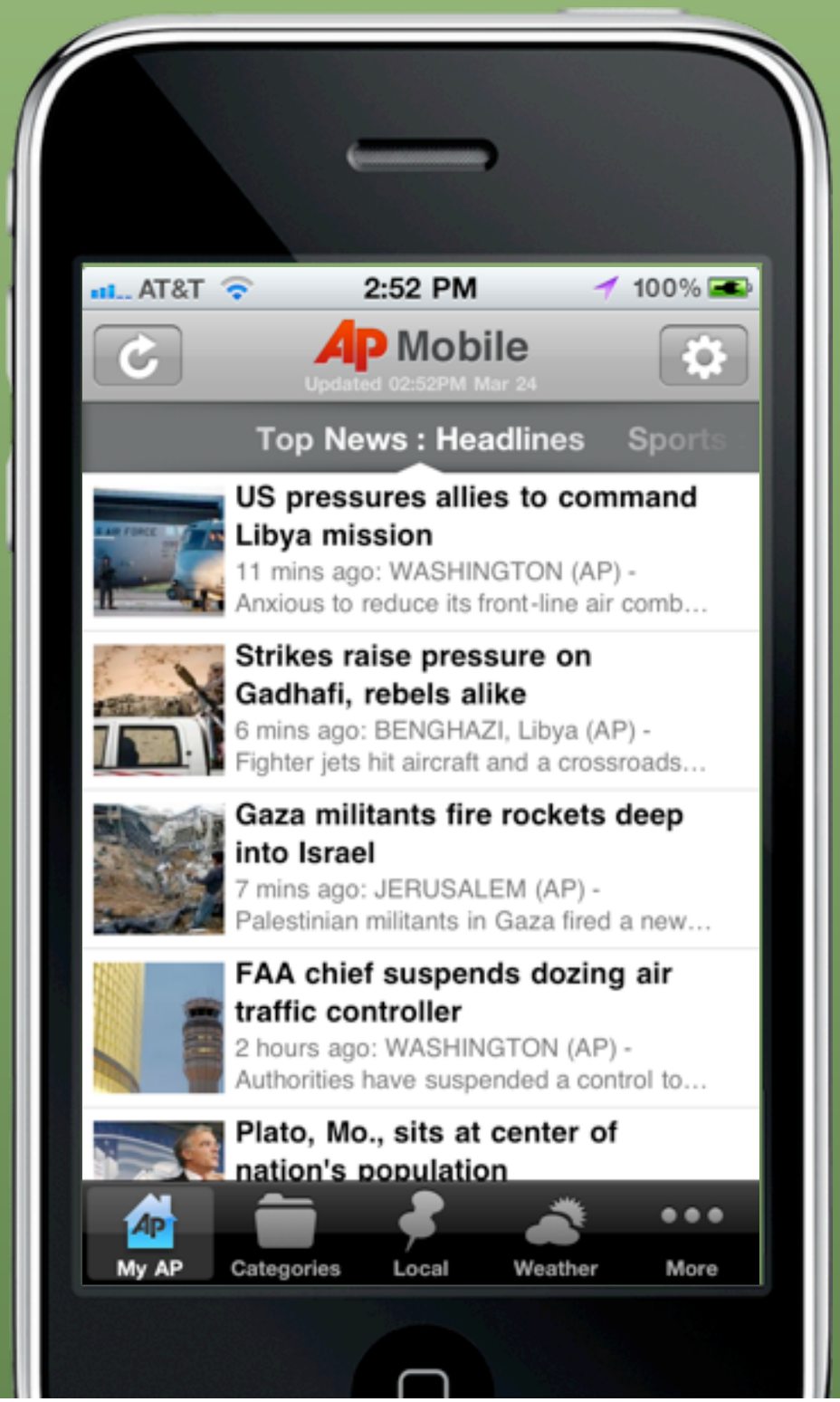
Nielsen Norman Group



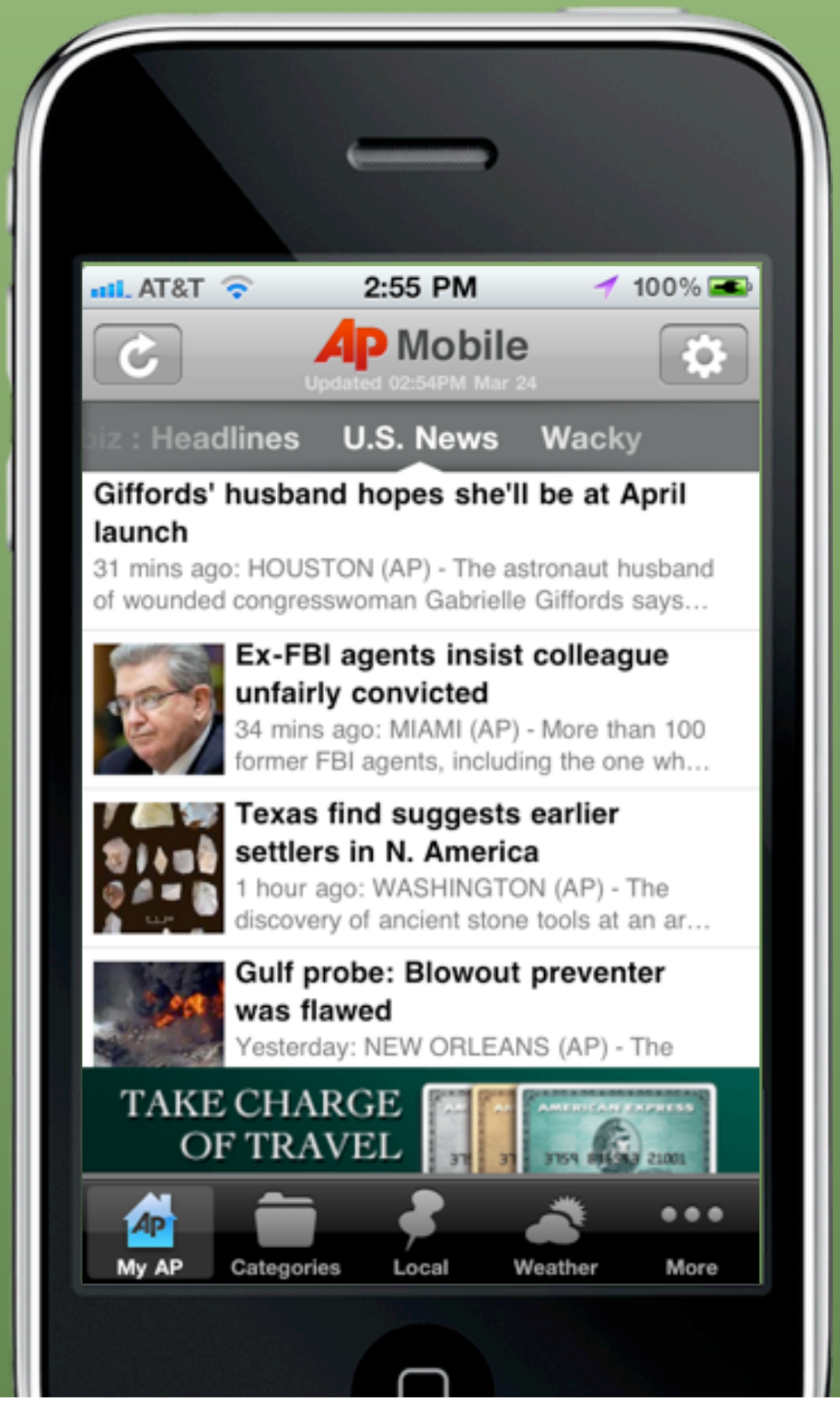
Cards vs Scroll



Cards vs Scroll



Cards vs Scroll



Cards vs Scroll



Cards vs Scroll



Cards vs Scroll



Affordances

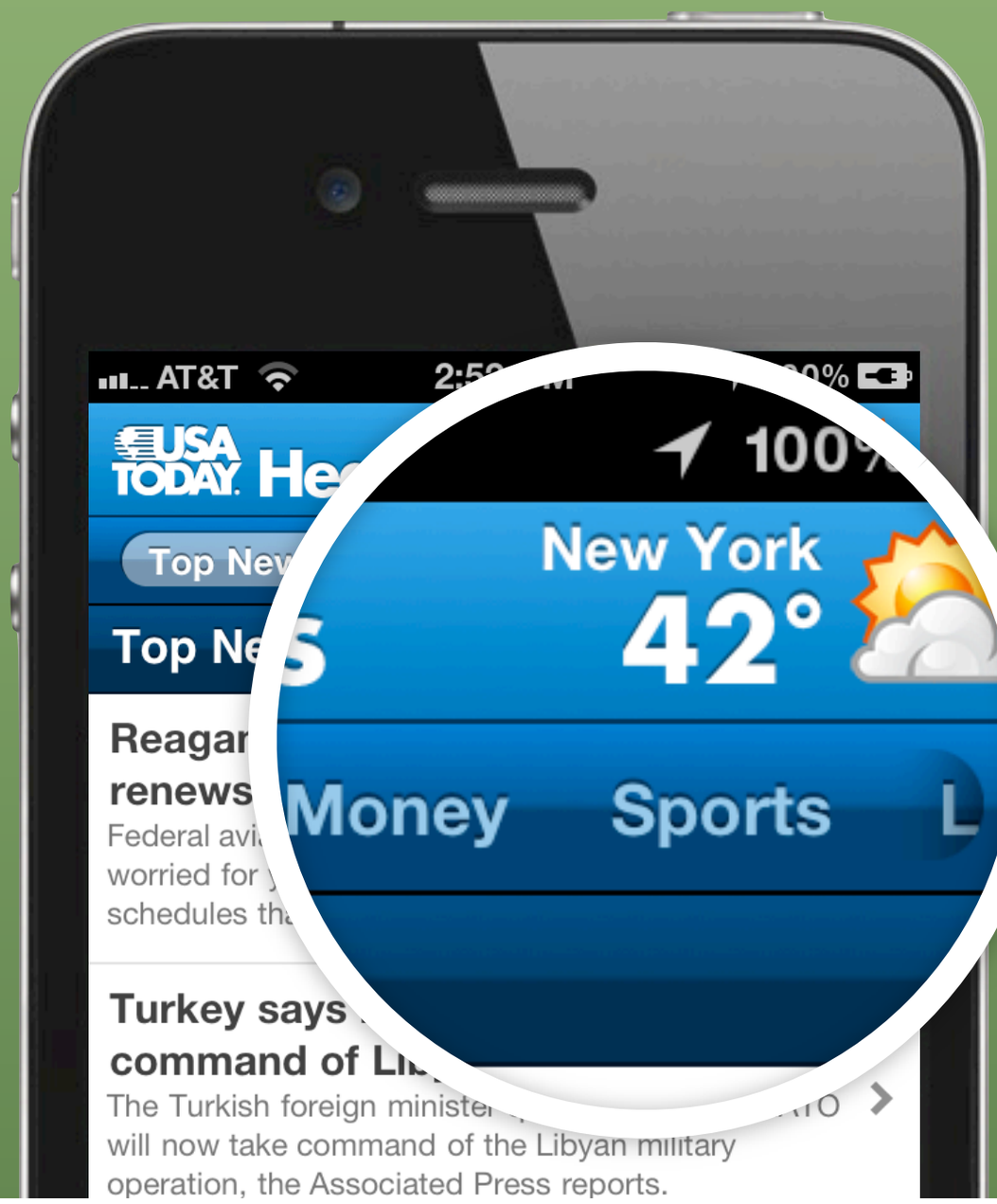
Be clear about what is clickable

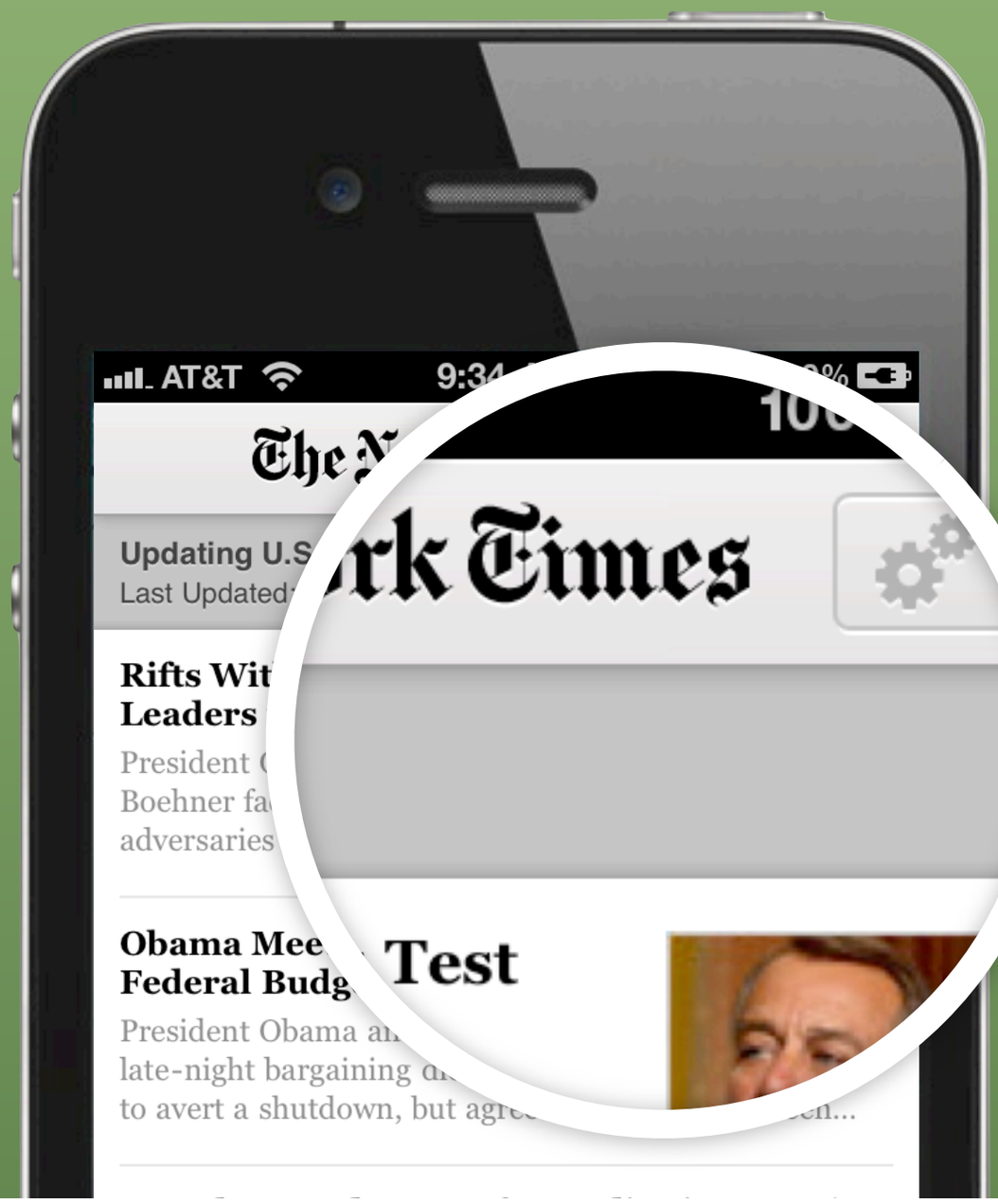




Affordances

Be clear about what is clickable





Affordances









